Parallel Computer Architecture

Lecture 5:

- Post Moore Ideas
  - **Neuromorphic Computing**
  - **Photonic Computing and Communication**

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Spring 2026

Outline

- Introduction
  - Background and Motivations
  - Problem Statement and Objectives
- Related Work
- Approach, Accomplishments, and Results
- Conclusion

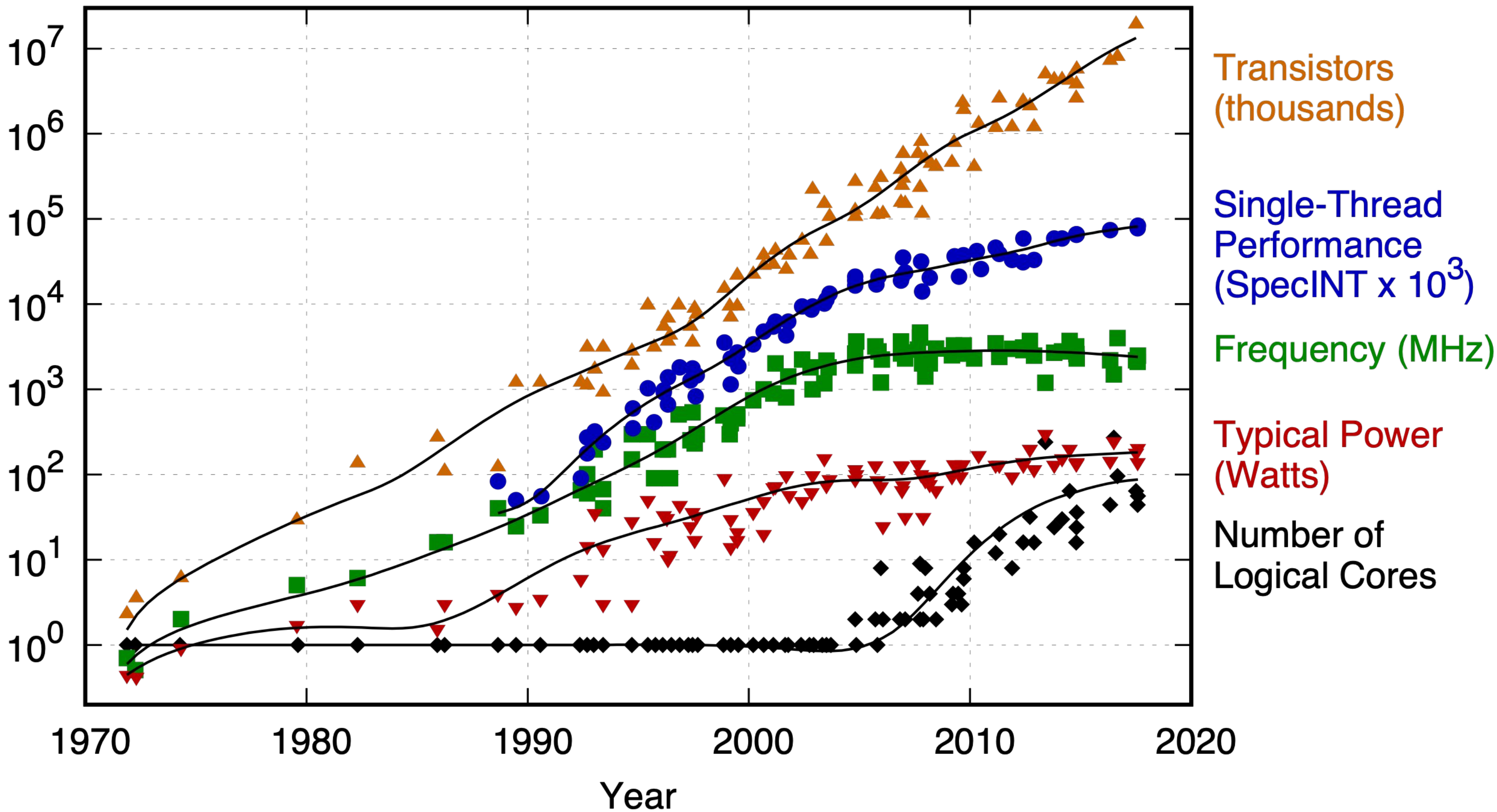
Background & Motivations
Computing Paradigms

Architecture / Technology	Von-Neumann	Neuromorphic	Quantum
Electronic			
Photonic		**This Research**	

Background & Motivations

End of Moore's Law

42 Years of Microprocessor Trend Data

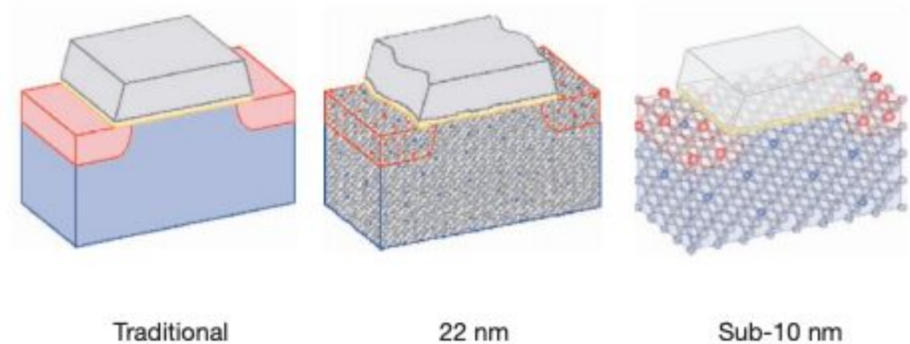


- Computing performance have been challenged by the end of Dennard scaling in 2004
- "Will need non-traditional computing models and accelerators, such as neuro-inspired and quantum accelerators"

Source: https://researchcomputingservices.github.io/parallel-computing/aio/ July 5th, 2020

Background & Motivations – Fundamental & Architectural Limitations of electronic computing

- Fundamental physics limitation for electronics
  - Components size approaching the atomic scale
  - Prone to excessive (and likely prohibitive) thermal noise
  - Quantum tunneling
  - Capacitive effects

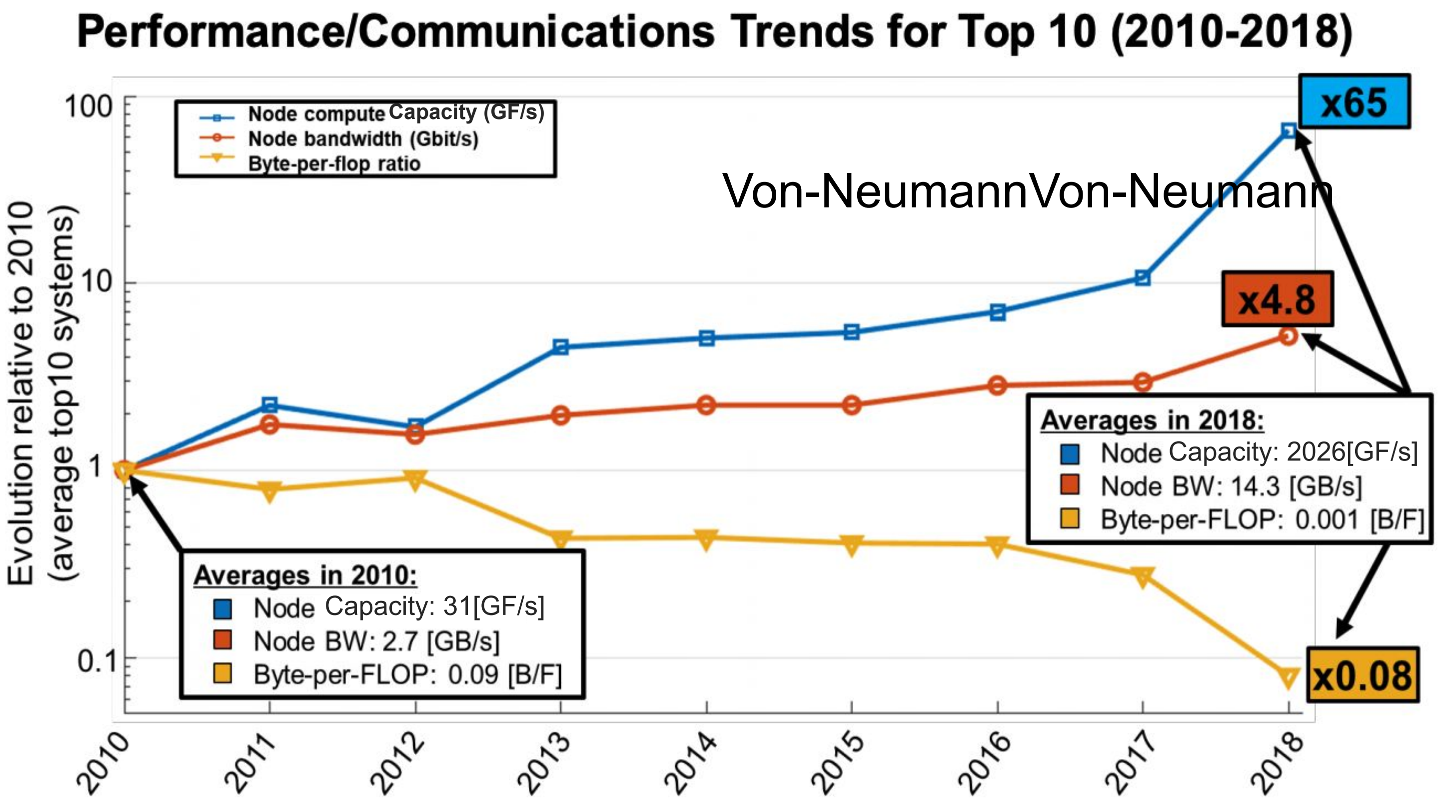


- Architectural limitation
  - Von-Neumann architecture – memory/compute separation data movement bottleneck
    - Bandwidth
    - Fan-out

Background & Motivations

Computation - Communication Disparity

- Node computation capacity increasing
- Node bandwidth increasing
- Disparity between the computation and communication bandwidth grows
- Photonics can address both



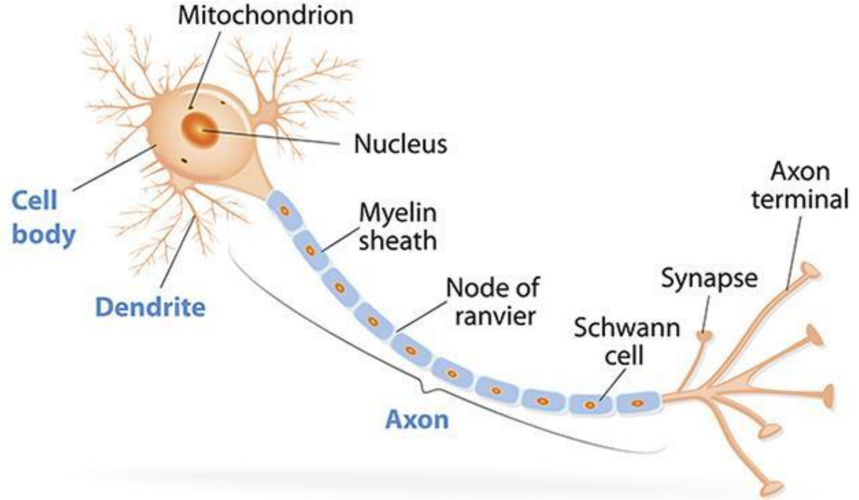
Sunway TaihuLight (Nov 2017)
B/F = 0.004

Summit HPC (June 2018)
B/F = 0.0005

Bergman, Keren. "Empowering Flexible and Scalable High Performance Architectures with Embedded Photonics." IPDPS. 2018.

Background & Motivations
Computational Fundamental

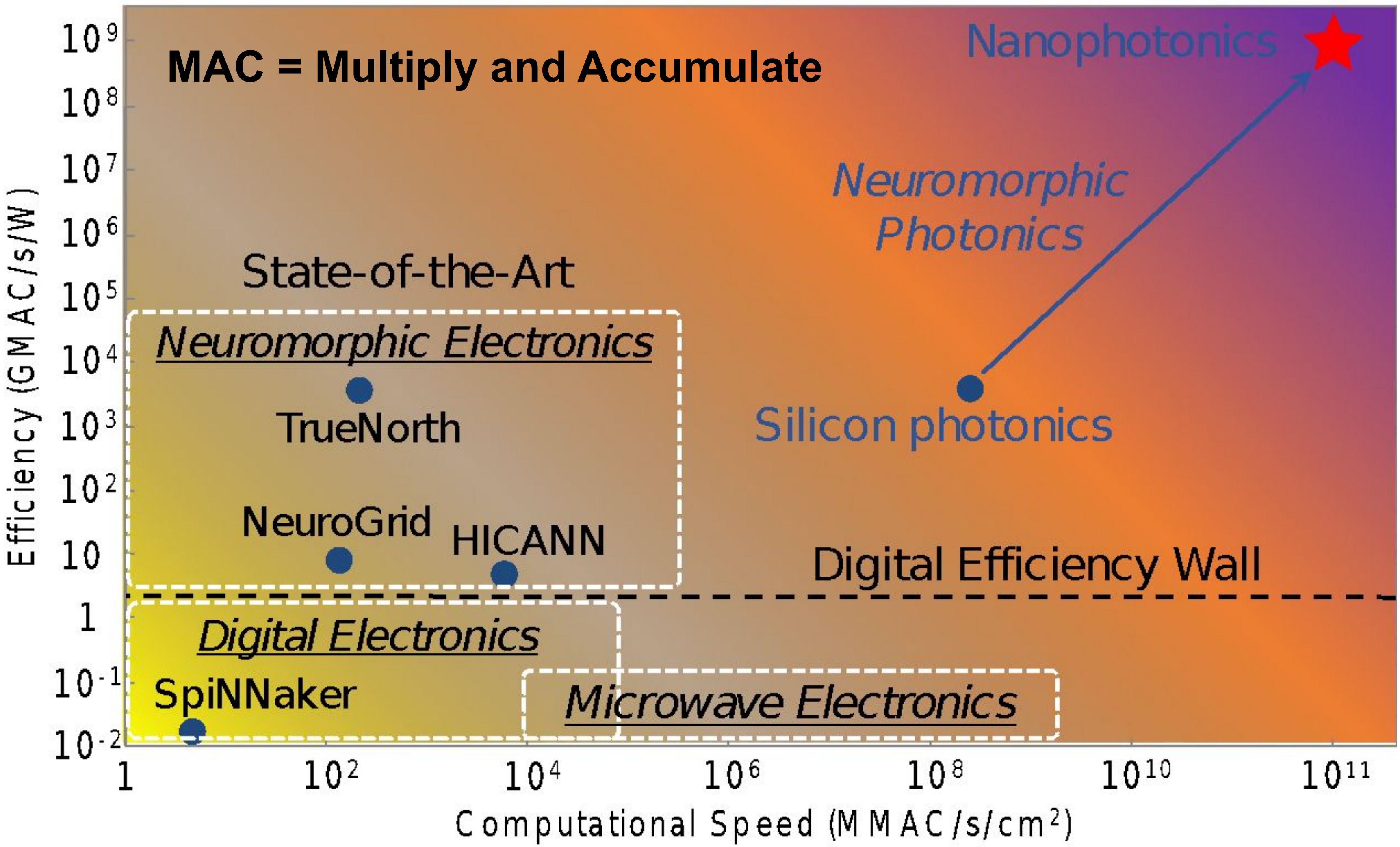
1. Processing/Memory proximity
   - Contrary to Von-Neumann arch
     - Expensive communication
     - Latency, Bandwidth, Energy constraints
   - Neuromorphic computing
     - Bringing memory close to computation
2. Heterogeneity in neurons
   - Sensory, Motor, Interneuron each with millions of variations
   - Specialized hardware (GMM accelerators, CNN accelerators)
3. Large fan-in/fan-out ~ 10^4 per neuron
   - Photonics offers better fan-in fan-out (WDM)
4. Long neurons
   - (optic nerve ~ 50 mm, motor nerves ~ up to 1 meters)
   - Photonics offer flat communication cost – World is my cache



1. https://faculty.washington.edu/chudler/facts.html
2. https://courses.lumenlearning.com/boundless-psychology/chapter/neurons/

Background & Motivations

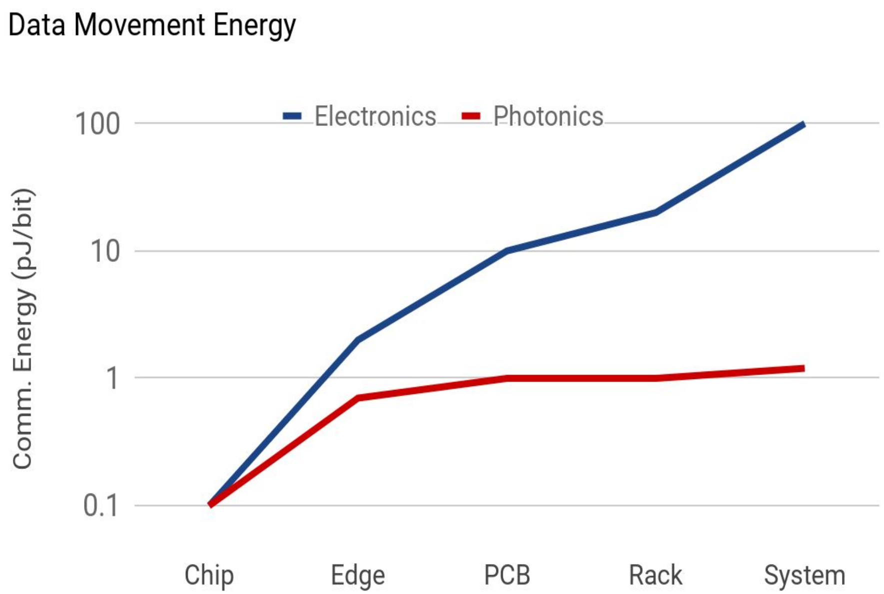
Potential for Photonic Neuromorphic

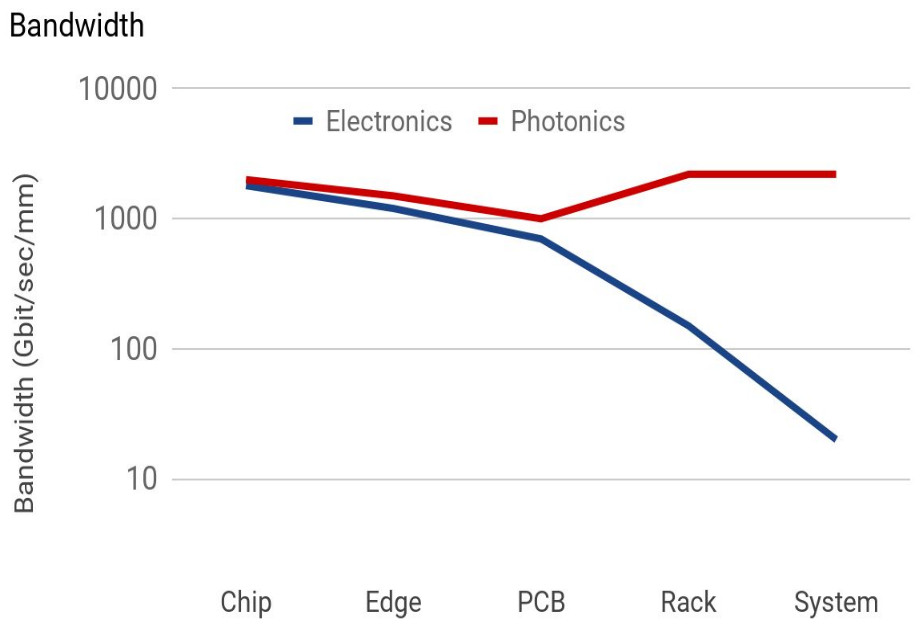


Prucnal & Shastri, *Neuromorphic Photonics* CRC Press, May 2017
Merolla et al. *Science* **345** (2014)
Furber et al. *Proceedings of the IEEE* **102** (2014)
Benjamin et al. *Proceedings of the IEEE* **102** (2014)

What Photonics bring to the table

- Distance-agnostic communication
- Flattens the Energy curve



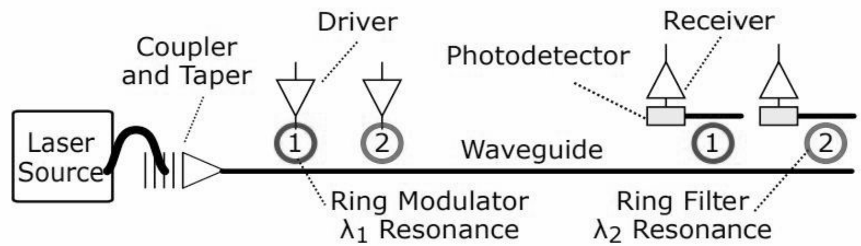


Cheng, Qixiang, et al. "Recent advances in optical technologies for data centers: a review." Optica 5.11 (2018): 1354-1370.

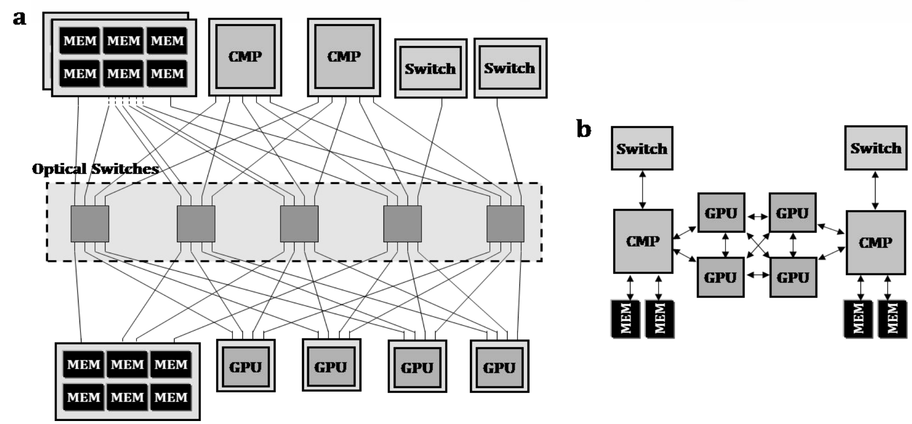
Background and Motivation

3 ways photonics can assist

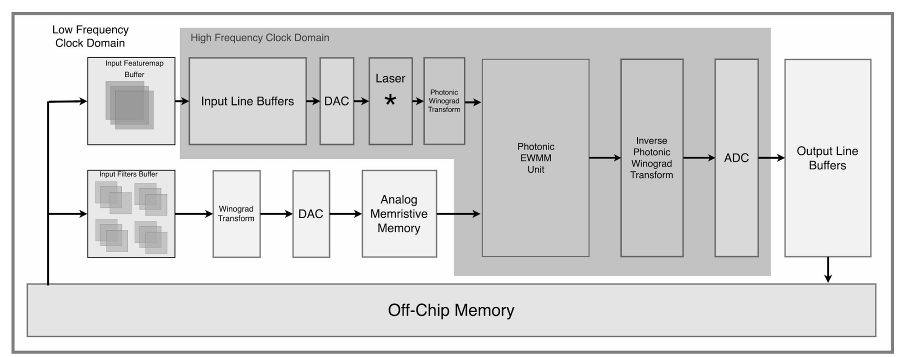
1. Photonic Links



2. Bandwidth Steering



3. Photonic Processors



Top Green 500 Energy Efficiency

- Energy efficiency is increasing from **June 2016** to **June 2018** very rapidly

June 2016

Name	Top500 rank	GFlop/W
Shoubu	94	6.7
Satsuki	486	6.2
Sunway TL	1	6.1

November 2016

Name	Top500 rank	GFlop/W
DGX Sat.V	28	9.5
Piz Daint	8	7.5
Shoubu	116	6.7
Sunway TL	1	6.1

June 2017

Name	Top500 rank	Gflop/W
TSUBAME3.0	61	14.1
kukai	465	14.0
AIST AI Cloud	148	12.7
RAIDEN	305	10.6
Wilkes-2	100	10.4
Piz Daint	3	10.4
Gyoukou	69	10.2
GOSAT-2	220	9.8
	31	9.5
DGX Sat.V	32	9.5
Reedbush-H	203	8.6
JADE	425	8.4
Cedar	86	8.0
DAVIDE	299	7.7
Shoubu	137	6.7
Hokule'a	466	6.7
Sunway TL	1	6.1

November 2017

Name	Top500 rank	GFlop/W
Shoubu B	259	17.0
Suiren2	307	16.8
Sakura	276	16.7
DGX Volta	149	15.1
Gyoukou	4	14.2
TSUBAME3.0	13	13.7
AIST AI Cloud	195	12.7
RAIDEN	419	10.6
Wilkes-2	115	10.4
Piz Daint	3	10.4
Reedbush-L	291	10.2
GOSAT-2	319	9.8
	35	9.5
DGX Saturn V	36	9.5
Era-AI	109	8.6
Reedbush-H	295	8.6
Cedar	94	8.0
DAVIDE	440	7.9
Shoubu	180	6.7
Sunway TL	1	6.1

June 2018

Name	Top 500	Gflop/W
Shoubu B	359	18.4
Suiren2	419	16.8
Sakura	385	16.7
DGX Volta	227	15.1
Summit	1	13.9
TSUBAME3.0	19	13.7
AIST AI Cloud	287	12.7
Sunway TL	2	6.1 (#23)

Background and Motivation

- Scaling performance under tight energy budget
  - Increase hit rates of close memory banks (Cache, DRAM, …)
  - Improve memory access energy efficiency
  - Improve data movement energy efficiency
  - Decrease data movement (Locality exploitation)
  - Increase computing energy efficiency

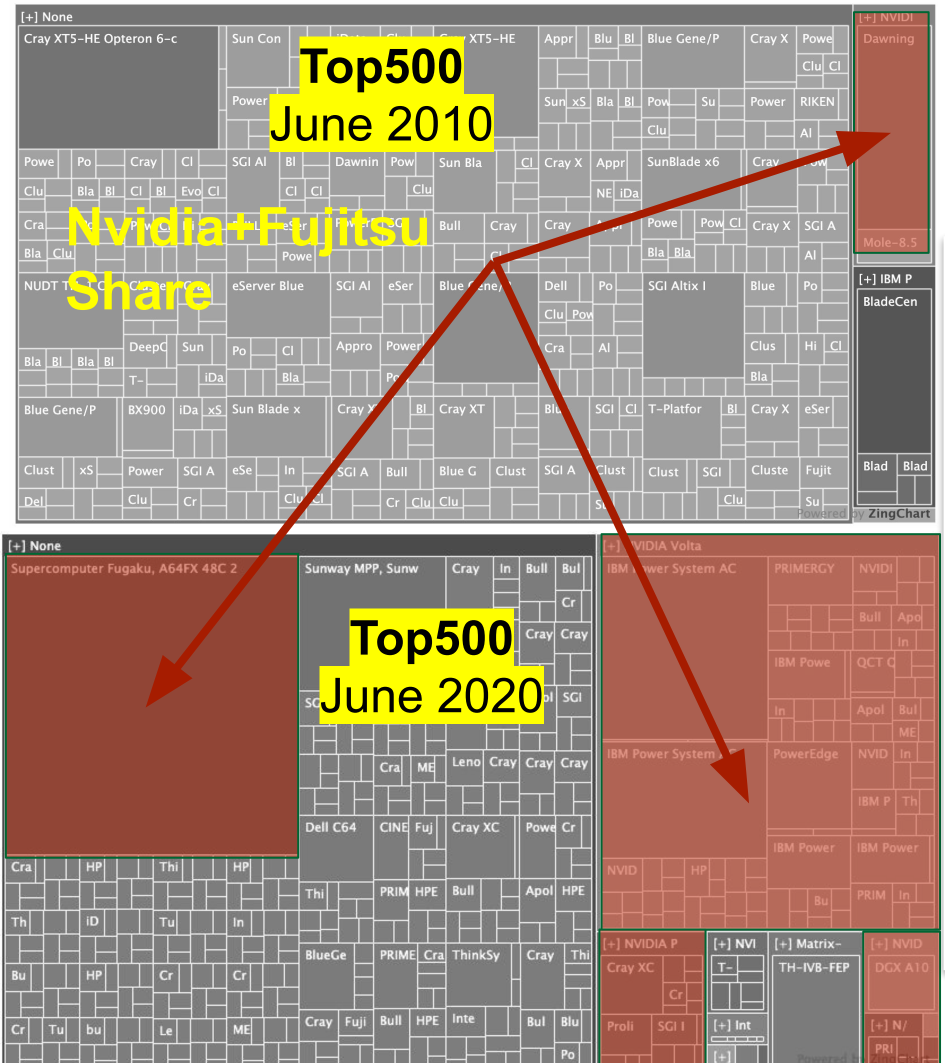
Activity	Energy
CMOS switching	***O**(10fJ)*
Single FLOP	***O**(10fJ)*
SRAM Access	***O**(100fJ)*
Across chip comm.	***O**(500fJ)*
Access DRAM	***O**(5pJ)*
Off-chip comm.	***O**(10pJ)*
Internet: optical WDM links	***O**(5nJ)*
Internet: routing	***O**(20nJ)*

Miller, David AB. "Attojoule optoelectronics for low-energy information processing and communications." Journal of Lightwave Technology 35.3 (2017): 346-396.

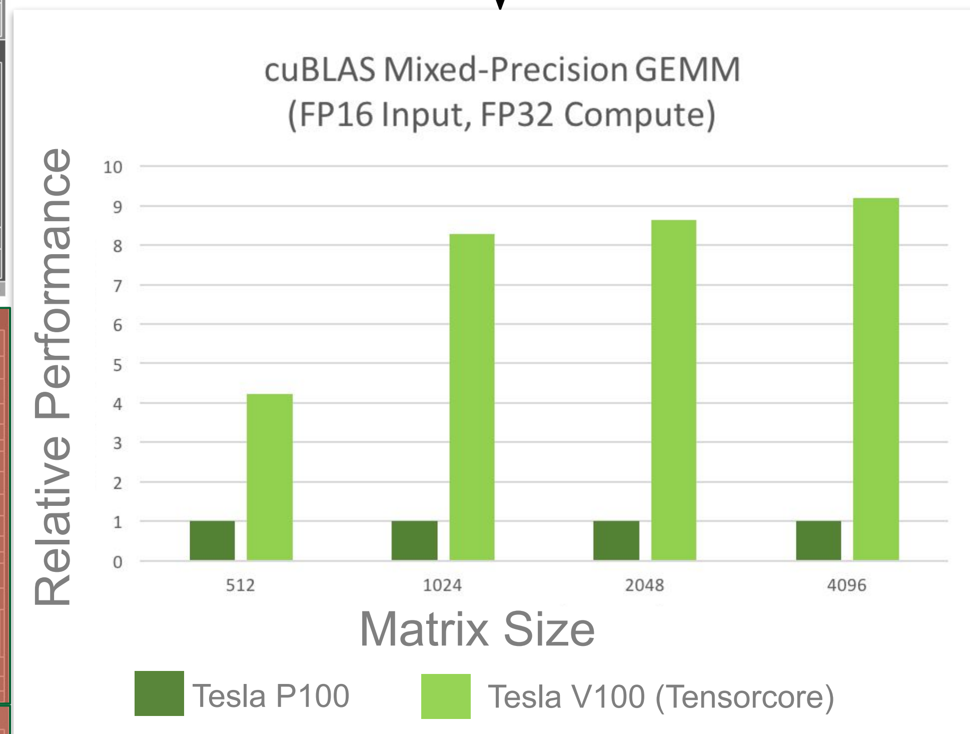
Background & Motivations

Machine Learning Workloads Dominating

- Workloads are changing in favor of Neural Networks



GPU with neural networks in mind **(Tensorcore)**



NVIDIA News Center Oct. 2017
https://news.developer.nvidia.com/programming-tensor-cores-in-cuda-9/

Motivation Summary

1. Electronics approaching limits
   - Optical advancements bring significant promise
   - Neuromorphic computing is a viable alternative to Von-Neumann architectures
2. Workloads are changing
   - Next generation tools must consider future workloads
   - I.E. data analytics, neural networks, etc.
3. Specialization in future HPC
   - No one size fits all solution
   - Moving from homogenous to heterogeneous

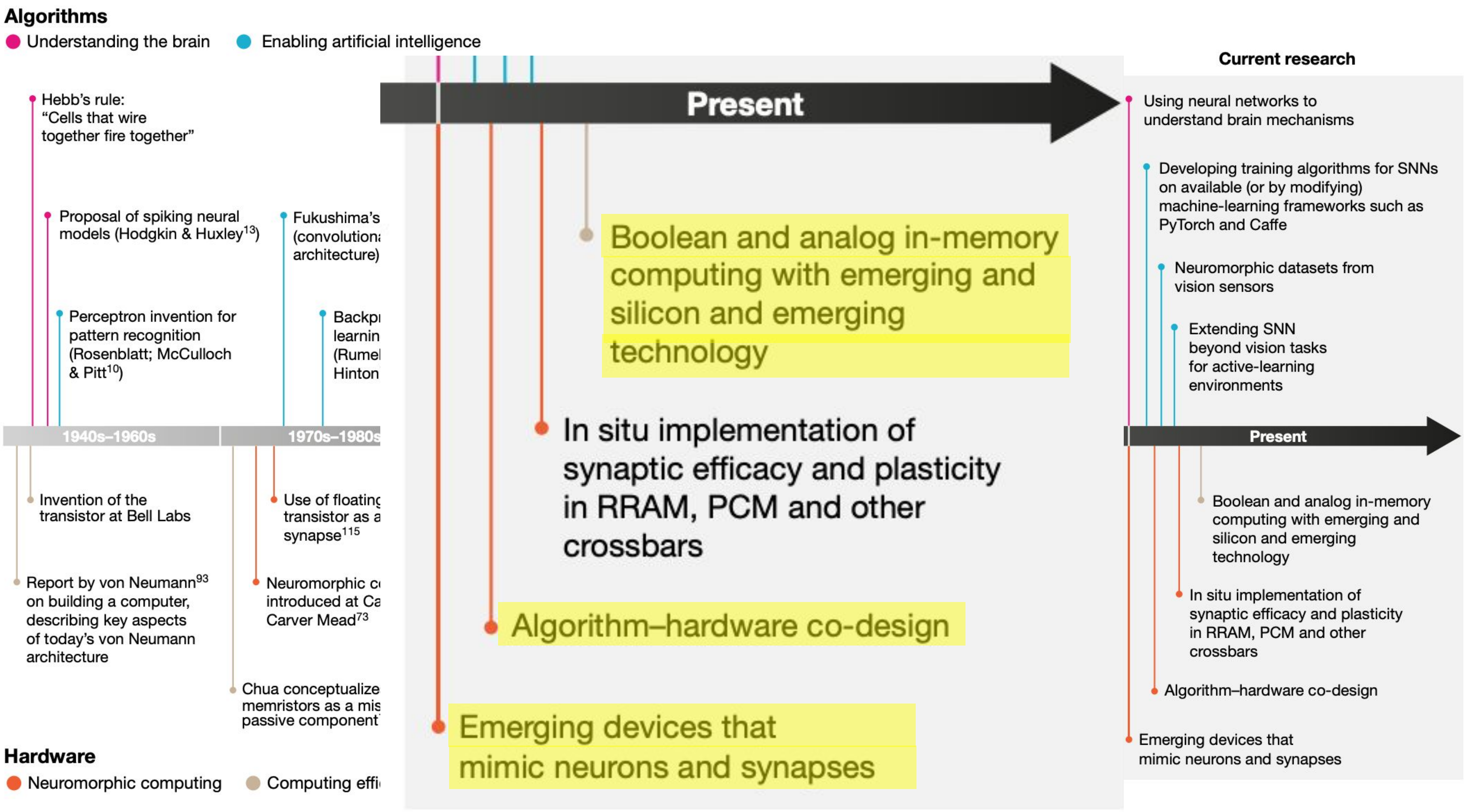
Problem Statement

Given (a) the advances in nanophononics and neuromorphics, which provide good energy efficiency (attojoule/MAC), high speed, and high bandwidth (WDM), and (b) whereas prior research work has merely been focused on device-level or small circuit- level neuromorphic nanophotonics;

1. Explore the design of ***efficient specialized neuromorphic photonic*** computing ***systems*** that
   - Can execute state-of-the-art neural networks
   - Exhibit great energy efficiency and speed
   - Are weight programmable
2. Explore the design space by
   - Highlighting potentials, pitfalls, and limitations
   - Developing a design and simulation framework for neuromorphic photonics

Related Work

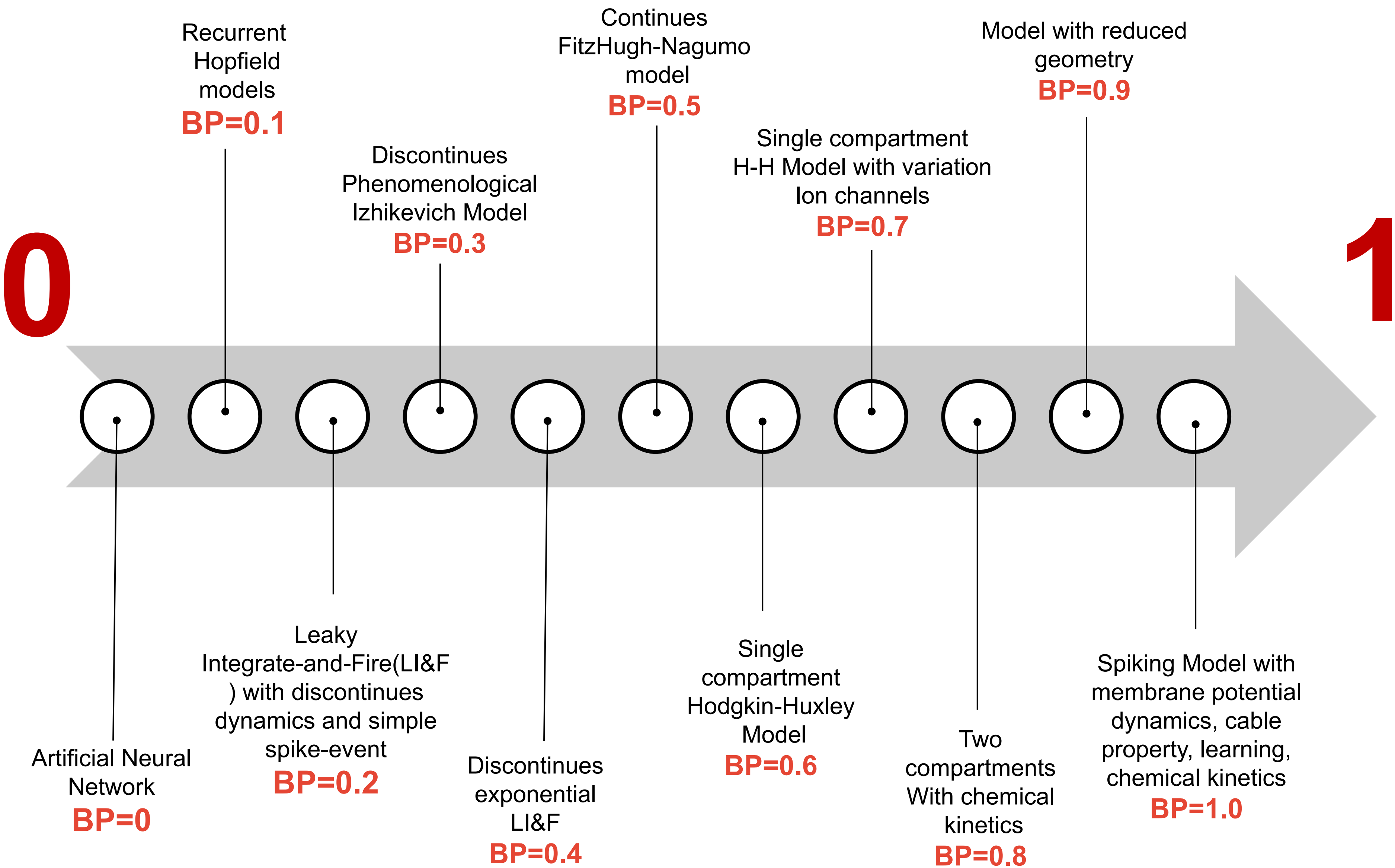
Evolution of Neuromorphic Computing



Roy K, Jaiswal A, Panda P. Towards spike-based machine intelligence with neuromorphic computing. Nature. 2019 Nov;575(7784):607-17.

Related Work

Biological Plausibility (BP)

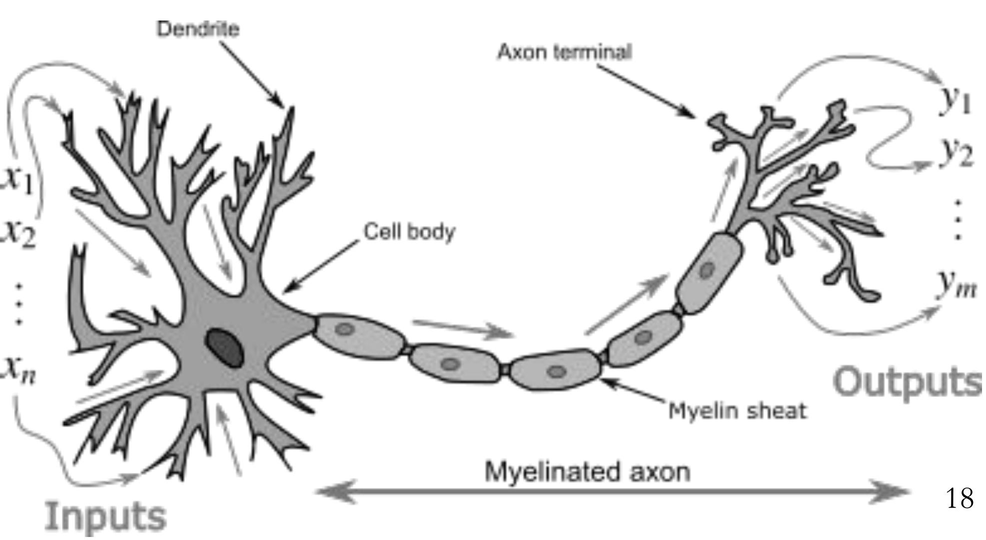


Related Work

Brain Neuron vs. Artificial Neuron

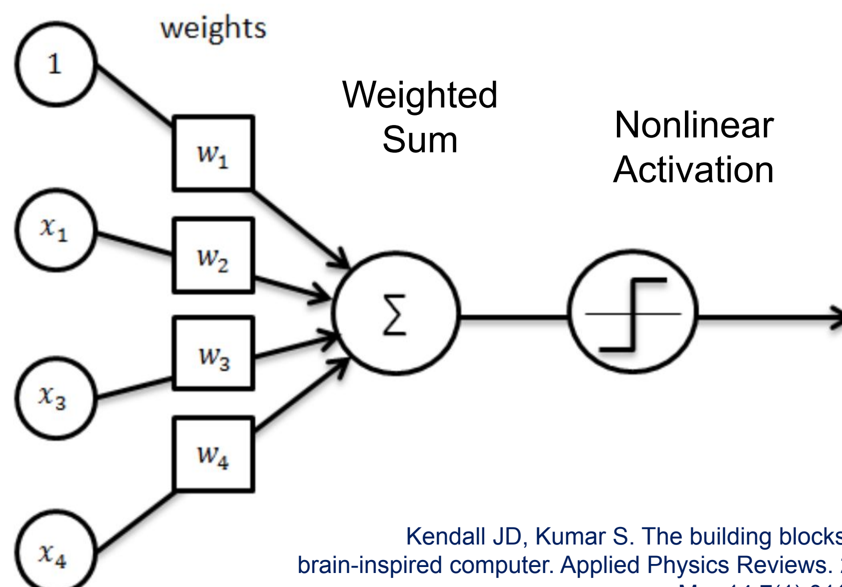
Neuron Cell

- Dendrites receive electrical signals from the axons of other neurons
- Electrical signals are modulated in various amounts between dendrites and axons
- Neuron fires an output signal only when the ***total*** strength of the input signals exceed a certain ***threshold***



Perceptron

- In perceptron, these electrical signals are represented as numerical values
- This is also modeled in the perceptron by multiplying each input value by a value called the weight
- The ***weighted sum*** and the ***nonlinear activation*** represents the aggregation and thresholding

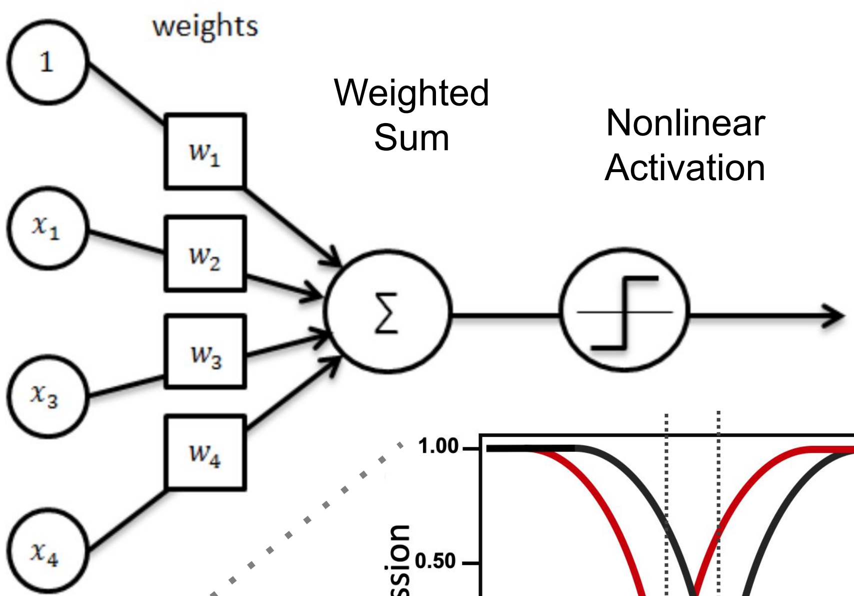


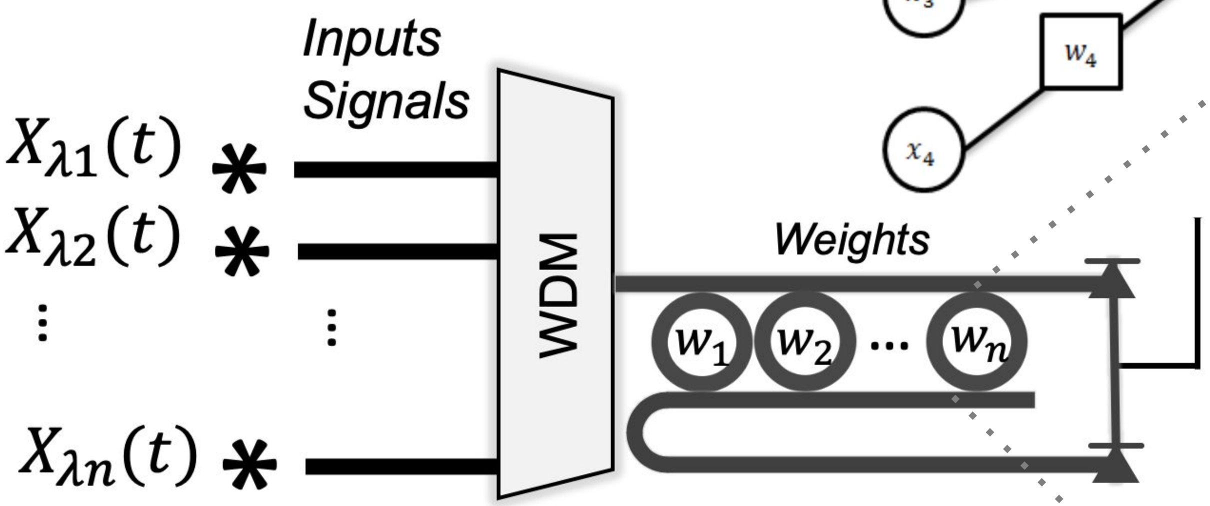
Kendall JD, Kumar S. The building blocks of a brain-inspired computer. Applied Physics Reviews. 2020 Mar 14;7(1):011305.

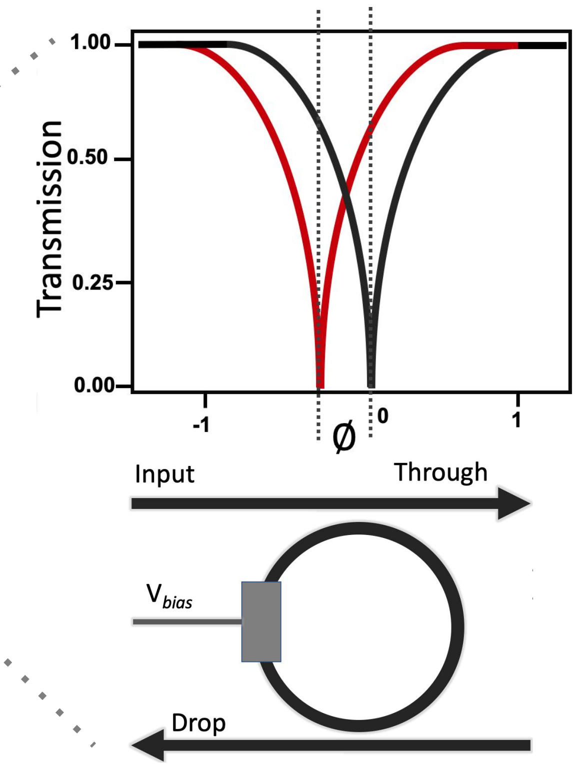
Photonic MAC Operator (Neuron)

MRR: Microring Resonator

- Photonic MAC using MRR
- Wavelength-Sensitive
- Can benefit from WDM



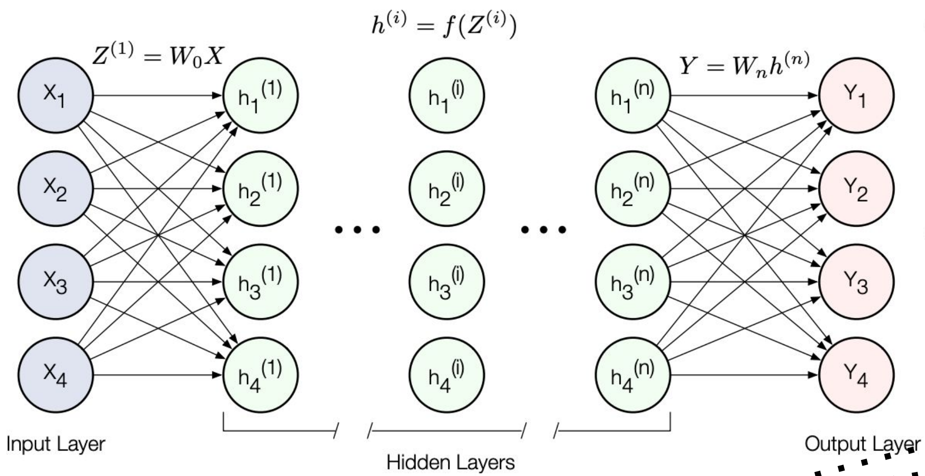


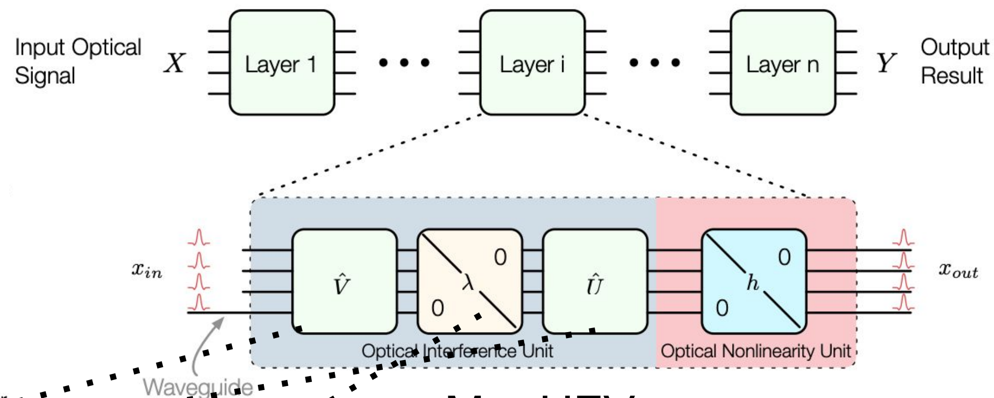


Tait AN, De Lima TF, Zhou E, Wu AX, Nahmias MA, Shastri BJ, Prucnal PR. Neuromorphic photonic networks using silicon photonic weight banks. Scientific reports. 2017 Aug 7;7(1):1-0.

Related Work

Coherent 2-layer Fully Connected Neural Network

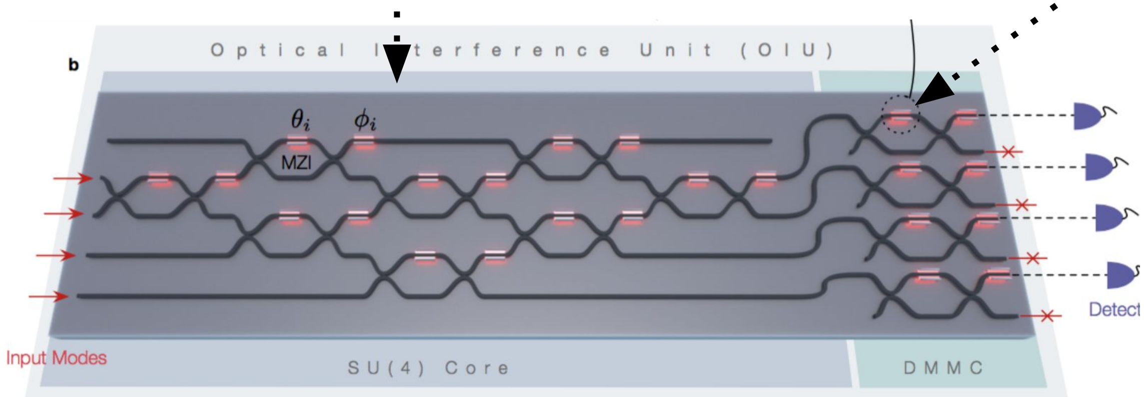


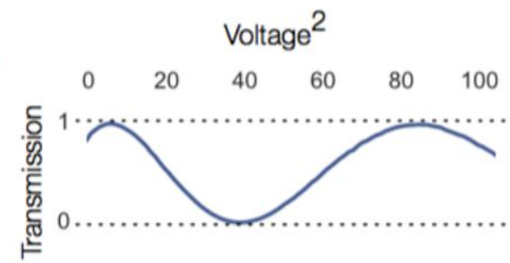


M = UΣV

V, U matrix parts

diagonal matrix multiplication core





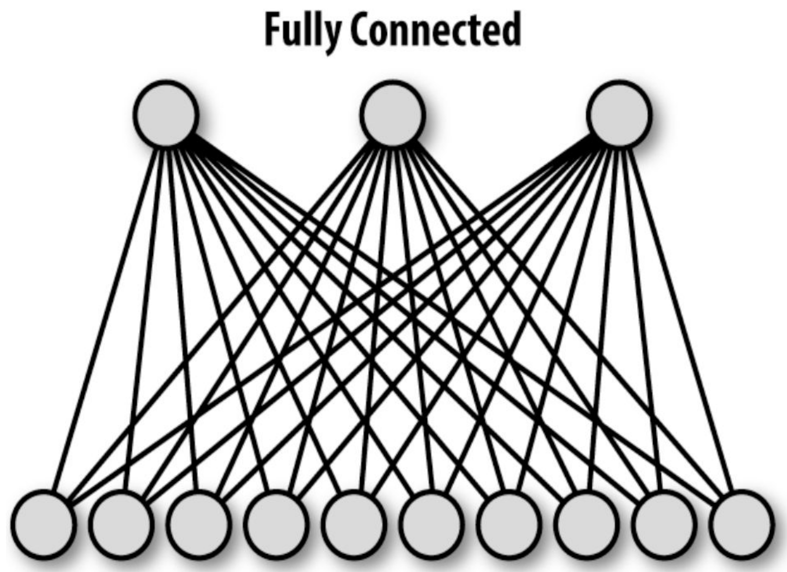
Shen, Yichen, et al. "Deep learning with coherent nanophotonic circuits." Nature Photonics 11.7 (2017): 441.

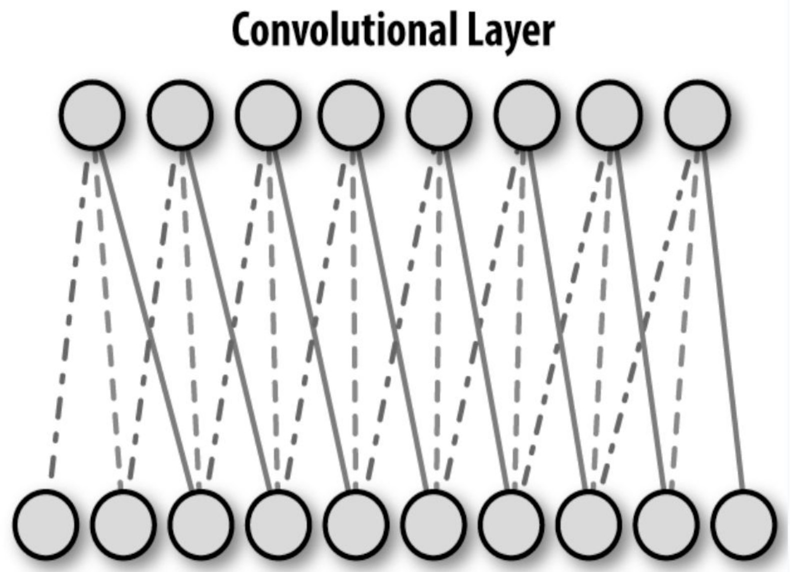
Approach

- We are seeking architecturally efficient neural network designs through
  - Specialization for the common cases
  - Use of transformations to reduce hardware
  - Leveraging novel devices
- Design specifications and constraints
  - Footprint: can be bulky - getting smaller
  - Power: extremely low power - length agnostic power
  - Speed: high speed
  - Bandwidth: WDM ~ 10 to 100 degree of parallelization
  - Photonic Memory ~ Research in progress, circuit switching
  - Interfacing: require convertors interfacing with electronics

Approach

Specialization: Fully-Connected vs. Convolutional Neural Network



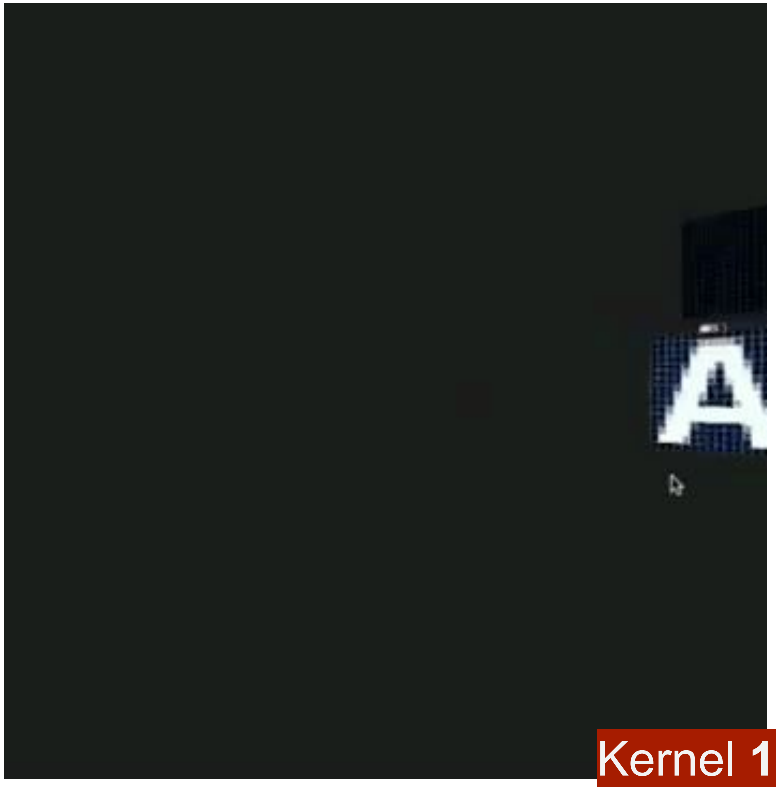


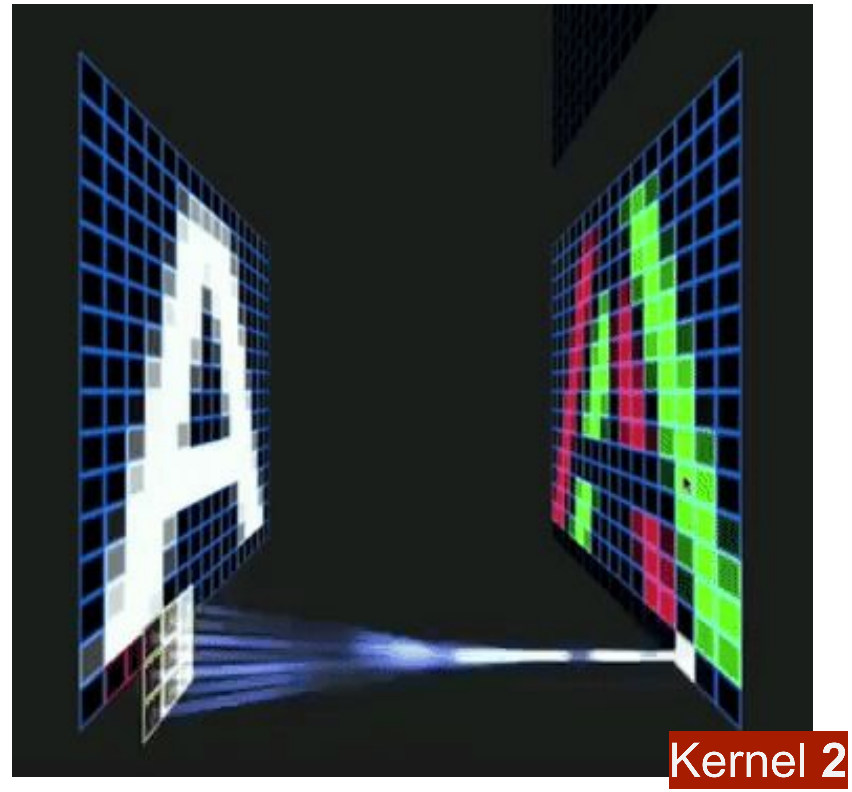
- All-to-all or most-to-most connections
- Not very practical on their own
- Hard to train
- Good with data with no local features - i.e., categorical

- Comprise bulk of the computations
- Almost all modern network architectures are CNNs, mostly comprised of CNNs, or have CNNs.
- Addressing most common applications with inexpensive hardware
- Good at feature detection
- Sparse connection between inputs and filters

Approach

Specialization: Parallelism Opportunities in CNNs



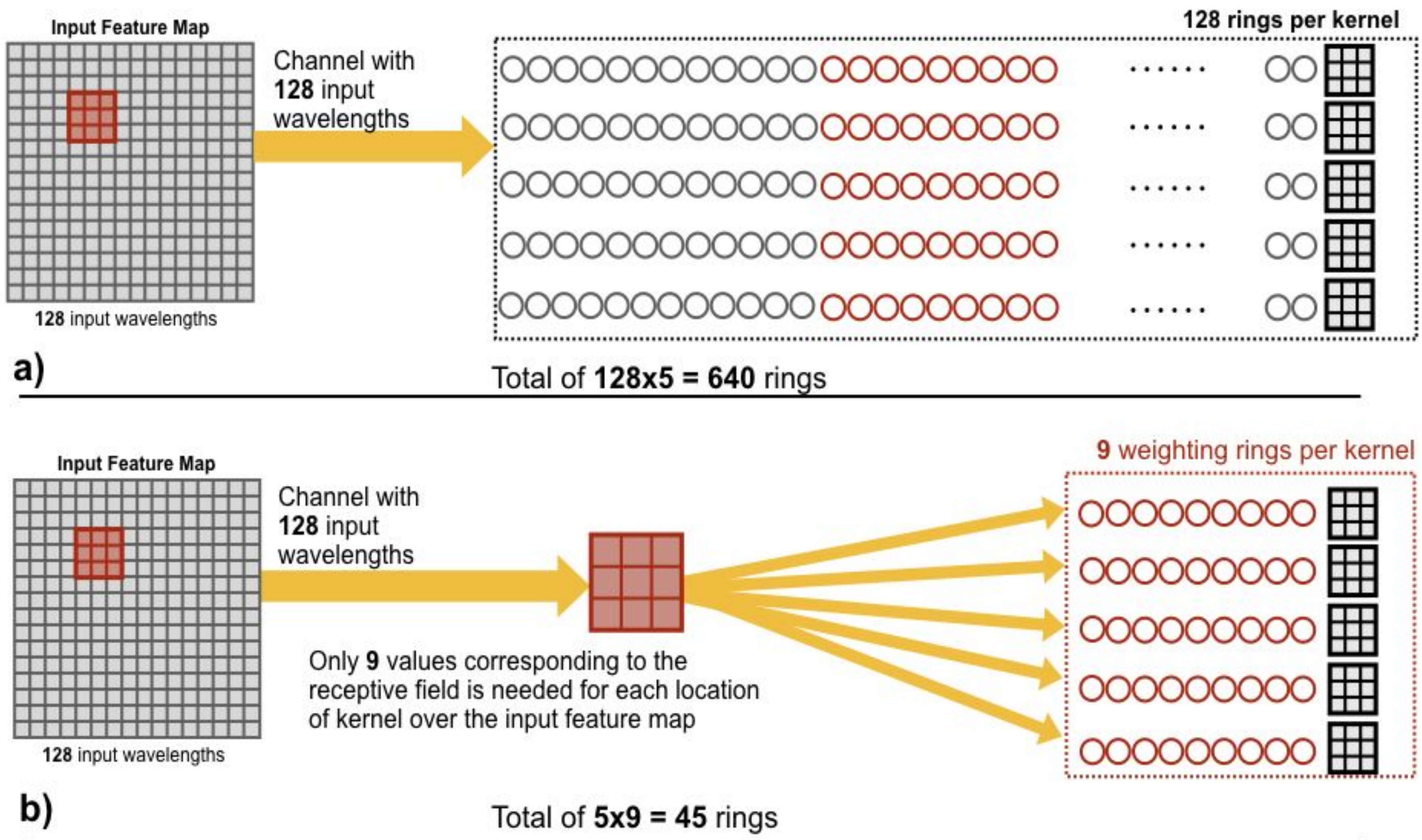


- A Kernel scans an input in search of a feature
- Multiple kernels (currently about hundreds) per layer of a CNN

Approach

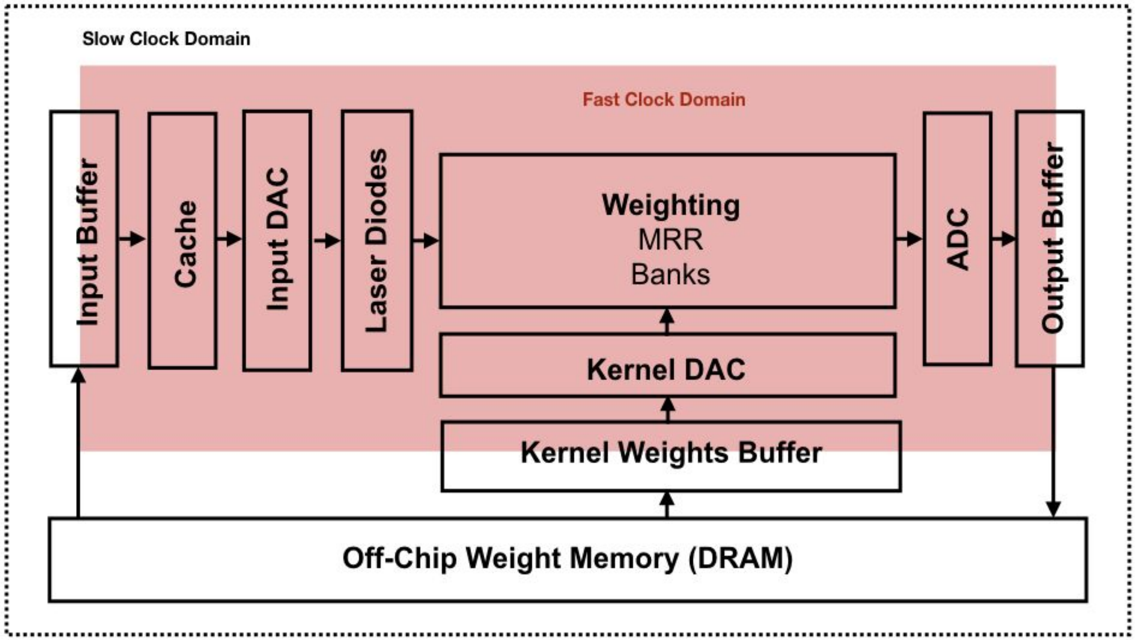
Specialization: Less Photonic Hardware with CNNs

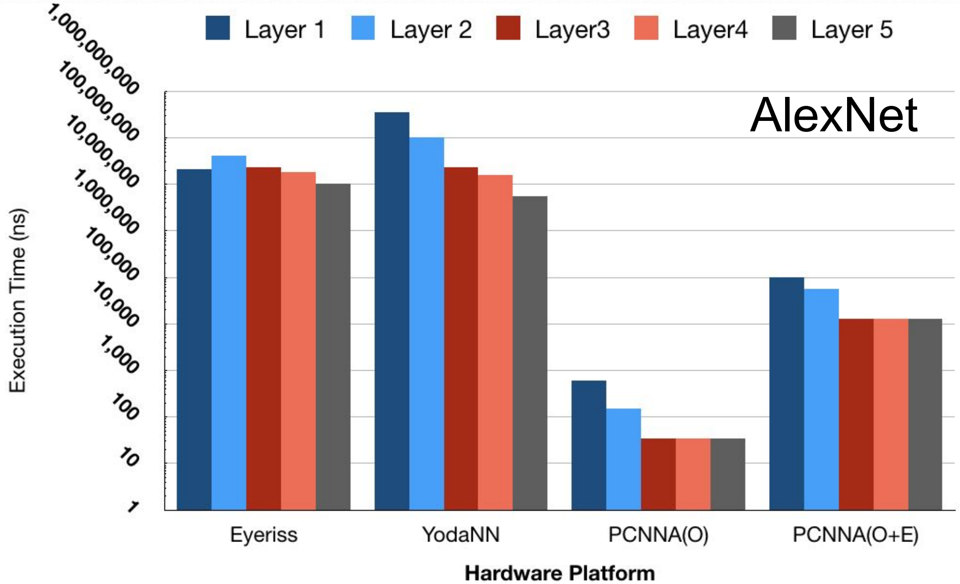
- Fewer MRRs make realization of these CNNs feasible



Specialization: Photonics Potential for CNNs - An Initial Study

- 10 ~ 100 WDM wavelengths can be used to map realize small kernel operations in CNN
- Potential of up to more than 3 orders of magnitude speedup in execution time
- The input DAC and output ADC conversions bound the speed





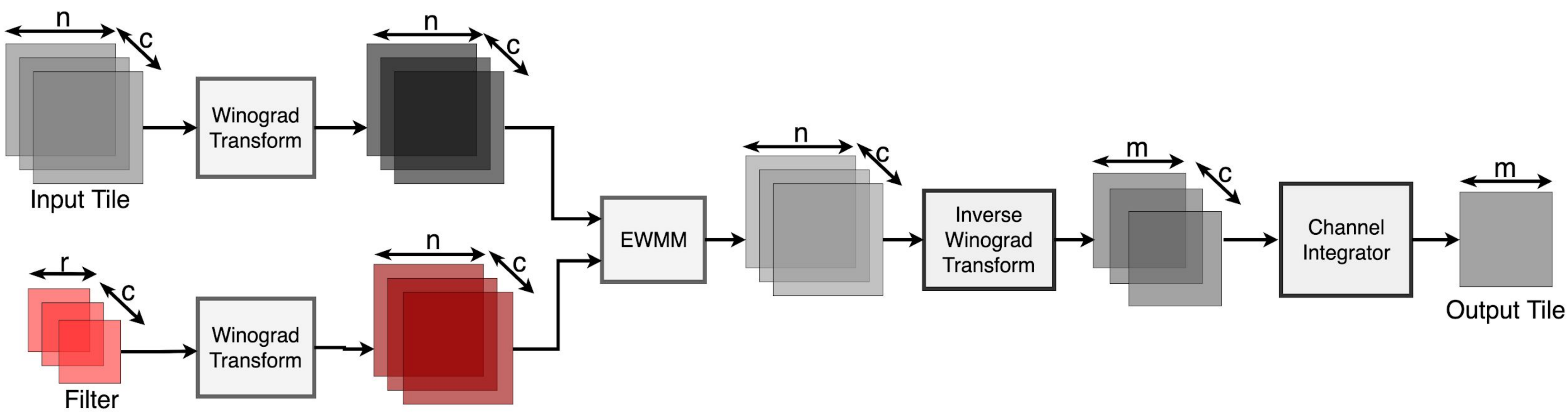
Mehrabian A, Al-Kabani Y, Sorger VJ, El-Ghazawi T. PCNNA: a photonic convolutional neural network accelerator. In2018 31st IEEE International System-on-Chip Conference (SOCC) 2018 Sep 4 (pp. 169-173). IEEE.

Approach
Transformations to Reduce Hardware Cost

1. General Matrix Multiplication (GEMM) approach
   - Toeplitz Matrix
   - Bloats matrix size More overhead
2. Fast Fourier Transform (FFT)
   - Better suited for larger kernel sizes
3. Winograd filtering algorithm
   - Better suited for smaller kernel sizes
   - Trades expensive multiplications for less expensive addition and accuracy

Winograd filtering algorithm is adopted to

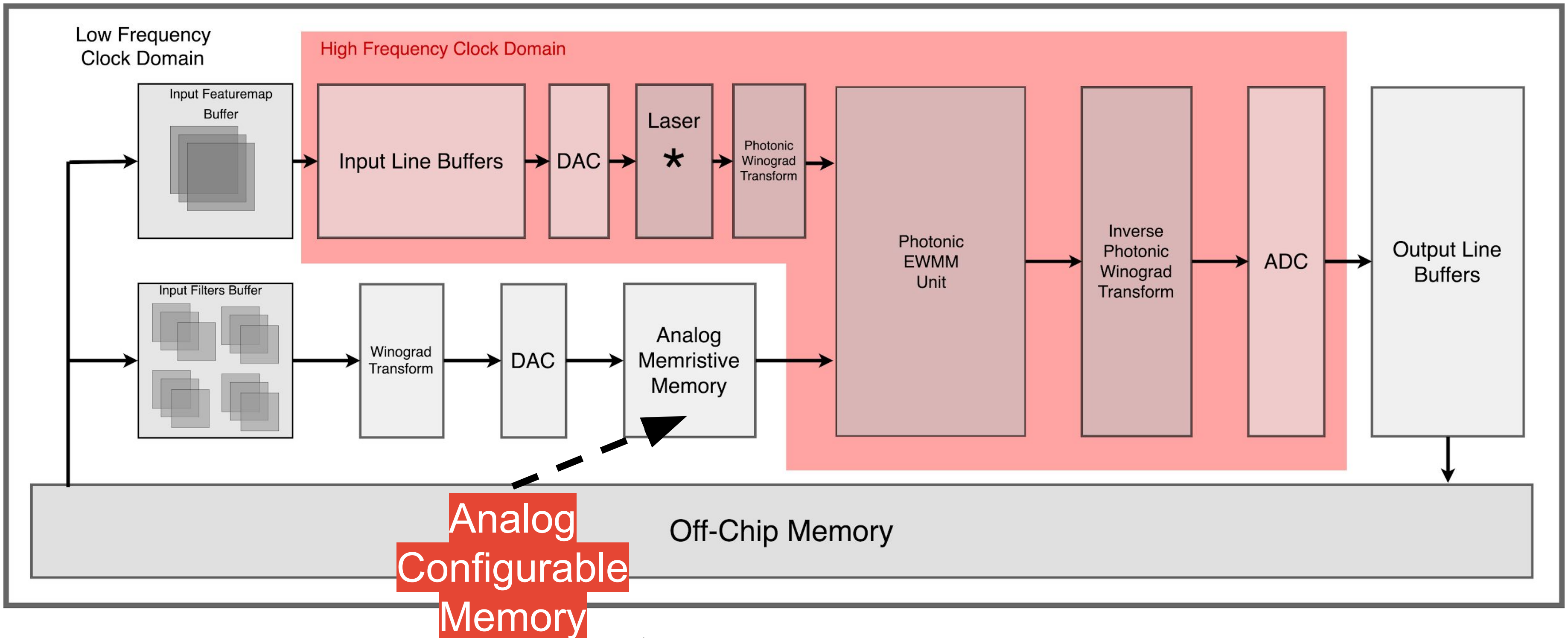
- Perform convolution
- Speedup the execution
- Reduce the computational complexity
- Reduce the number of multiplications at the cost of increased additions



Accomplishment

A CNN Architecture Accelerator

Inputs Path



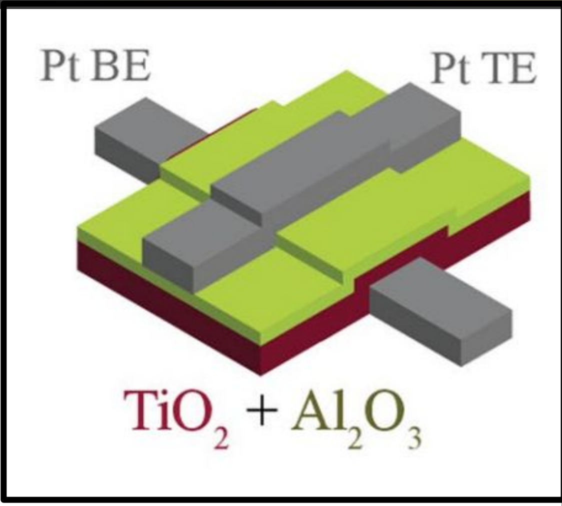
Weights Path

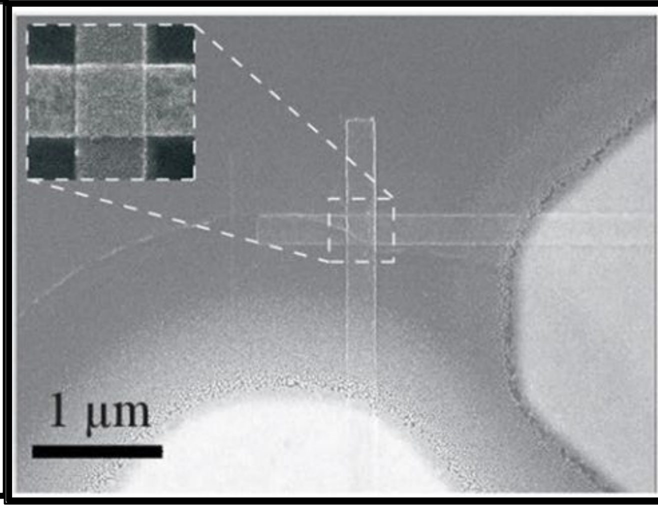
Mehrabian A, Miscuglio M, Alkabani Y, Sorger VJ, El-Ghazawi T. A Winograd-based Integrated Photonics Accelerator for Convolutional Neural Networks. IEEE Journal of Selected Topics in Quantum Electronics. 2019 Dec 3;26(1):1-2.

Mehrabian, A., Sorger, V. J., El-Ghazawi, T., & Miscuglio, M. (2020). U.S. Patent Application No. 16/507,854. - **Pending**

Approach

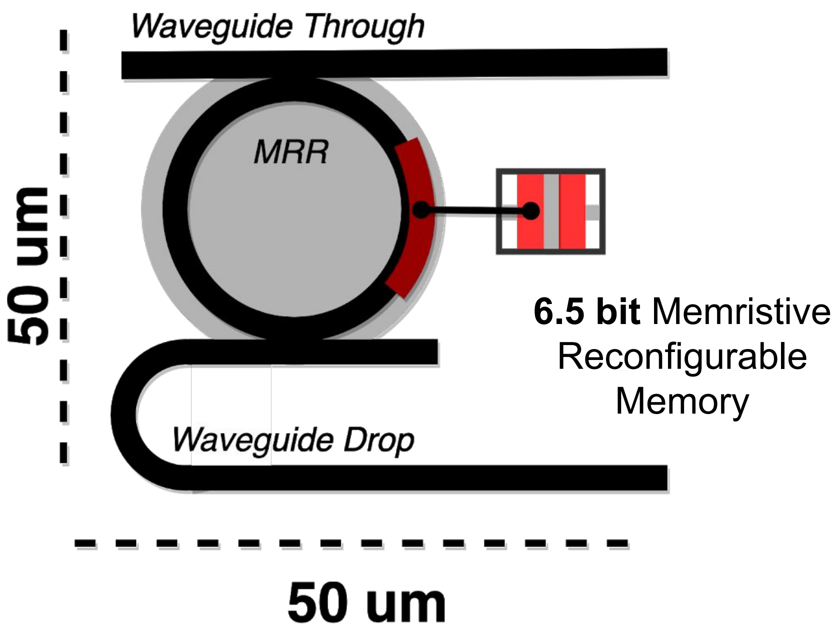
Novel Devices: Memristors as Analog Memory



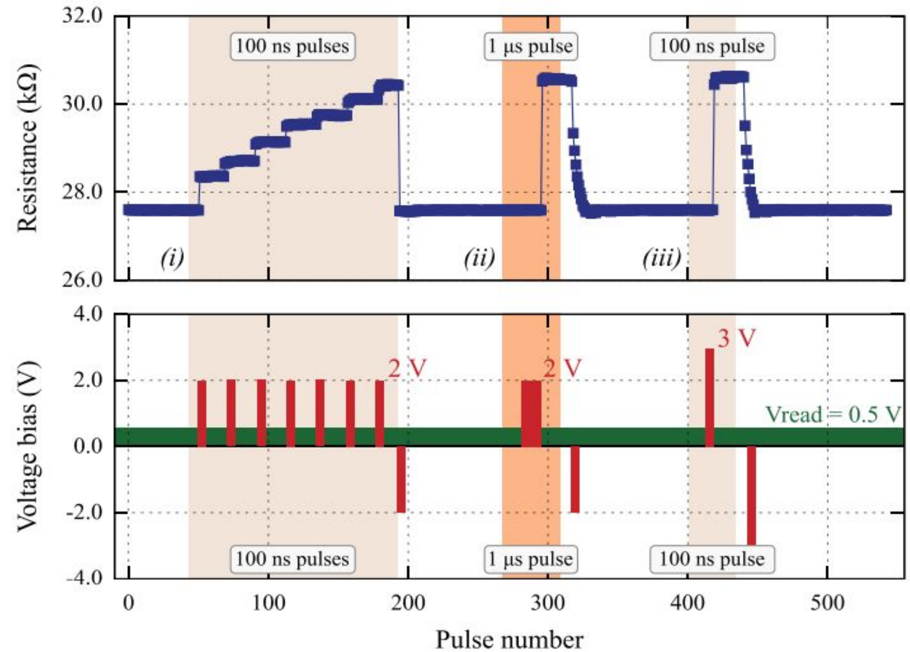


programming,
by modulating,

1. the number of pulses
2. the duration of the pulses or
3. the amplitude of the pulse.

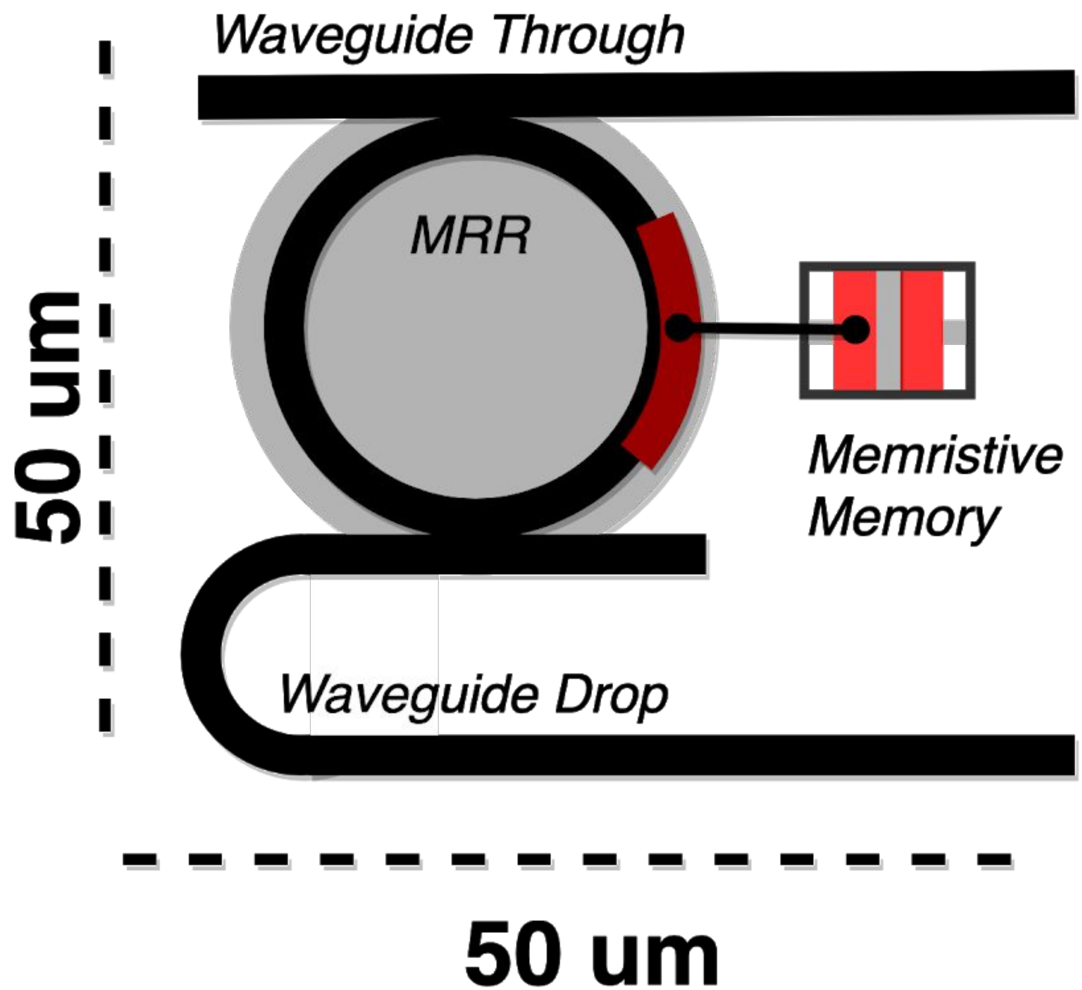


6.5 bit Memristive Reconfigurable Memory

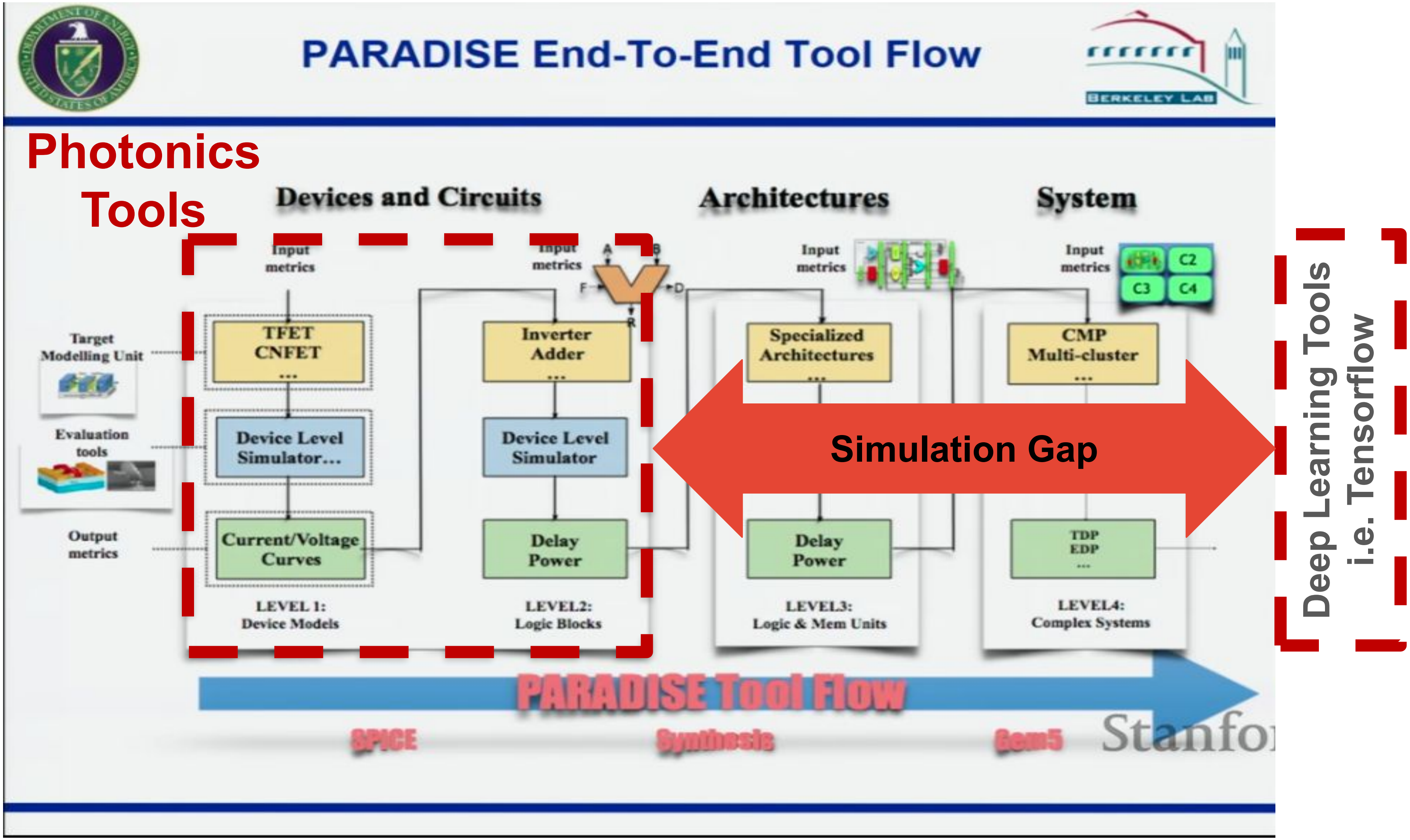


Stathopoulos S, Khiat A, Trapatseli M, Cortese S, Serb A, Valov I, Prodromakis T. Multibit memory operation of metal-oxide bi-layer memristors. Scientific reports. 2017 Dec 13;7(1):1-7.

MRR + Memristor Memory

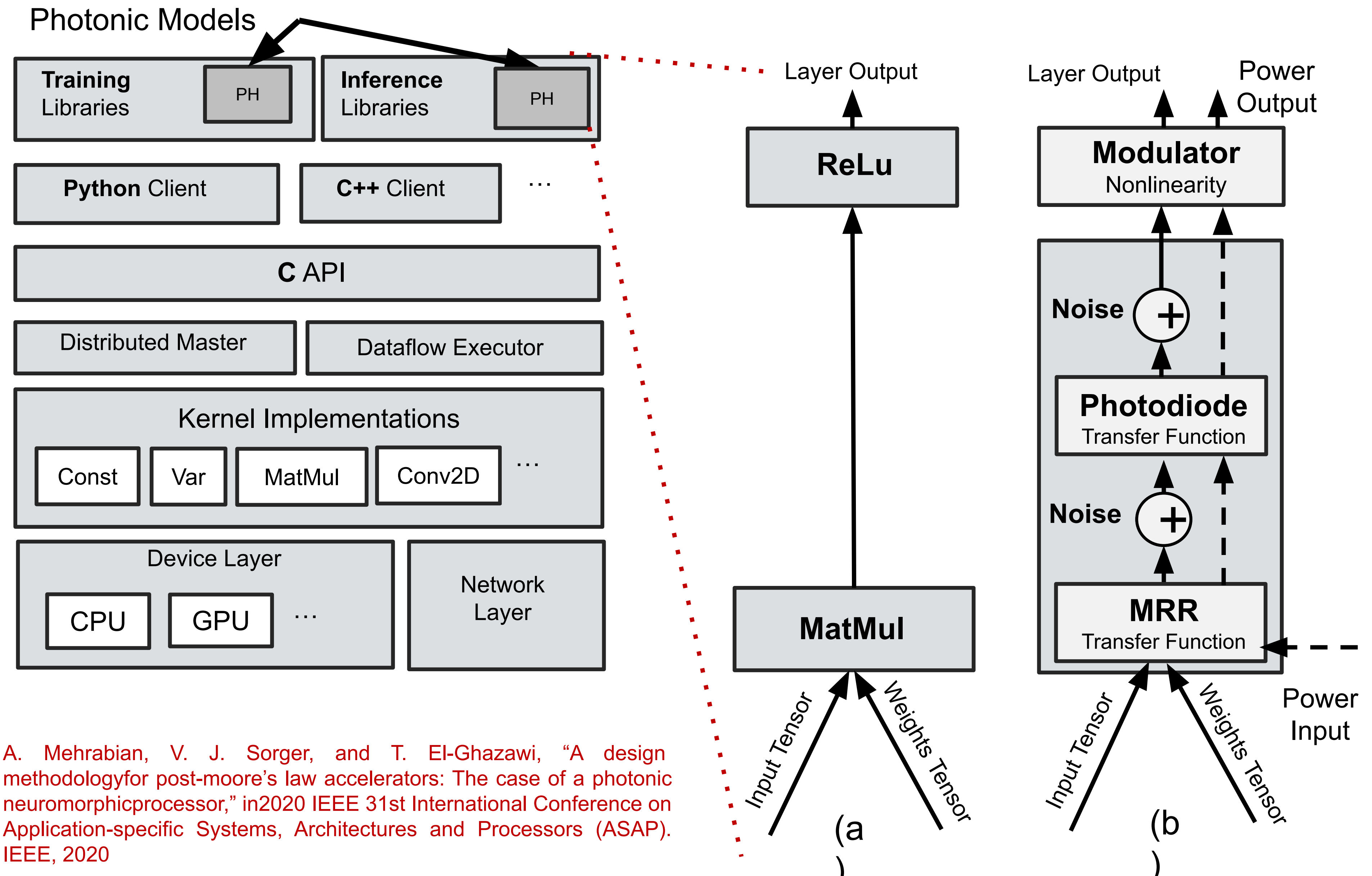


Approach
Lack of Design and Simulation Tools



Computation Research Conference 2018 Dr. George Michelogiannakis, Lawrence Berkeley National Laboratory

Accomplishment – A Design Methodology and Tool for Design and Simulation of Photonic Neural Networks

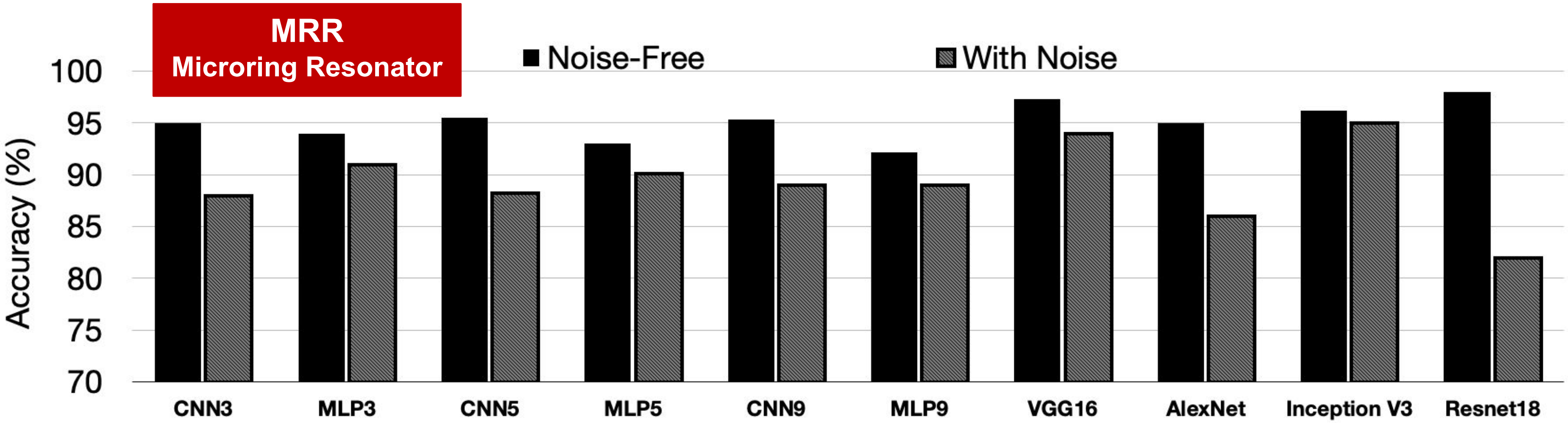


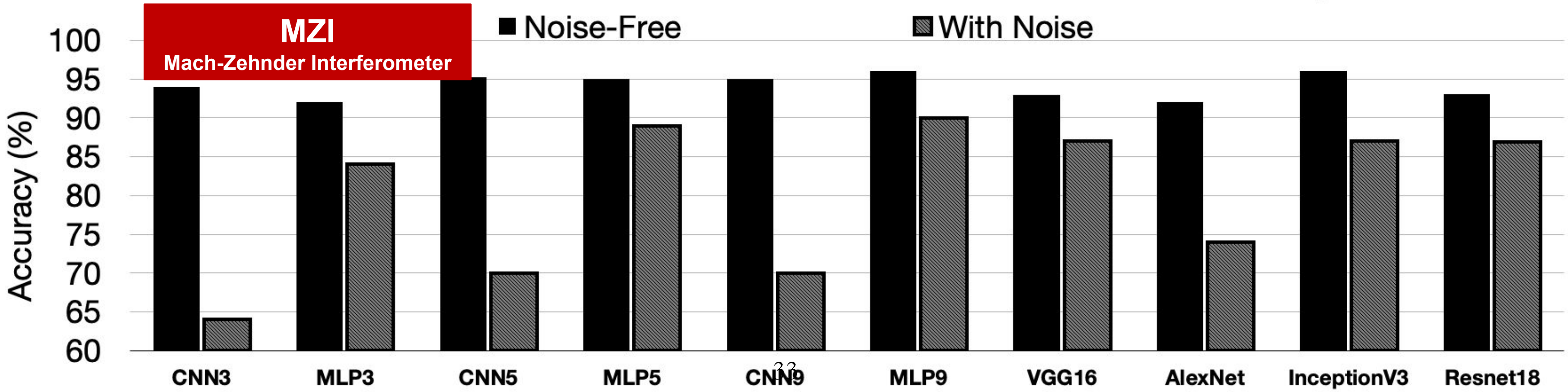
A. Mehrabian, V. J. Sorger, and T. El-Ghazawi, "A design methodologyfor post-moore's law accelerators: The case of a photonic neuromorphicprocessor," in2020 IEEE 31st International Conference on Application-specific Systems, Architectures and Processors (ASAP). IEEE, 2020

Results

Design Space Explorations

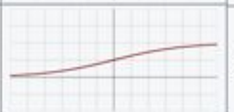
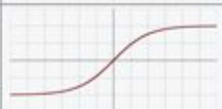
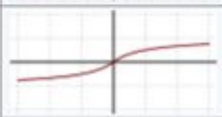
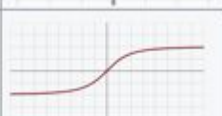
- Benchmark the performance of photonic implementations of state-of-the-art neural network
- Compare their realization with different photonic devices, i.e. MRR, MZI





Preliminary Results – Design Space Explorations –

Examples of Commonly Used Nonlinear Activation Functions

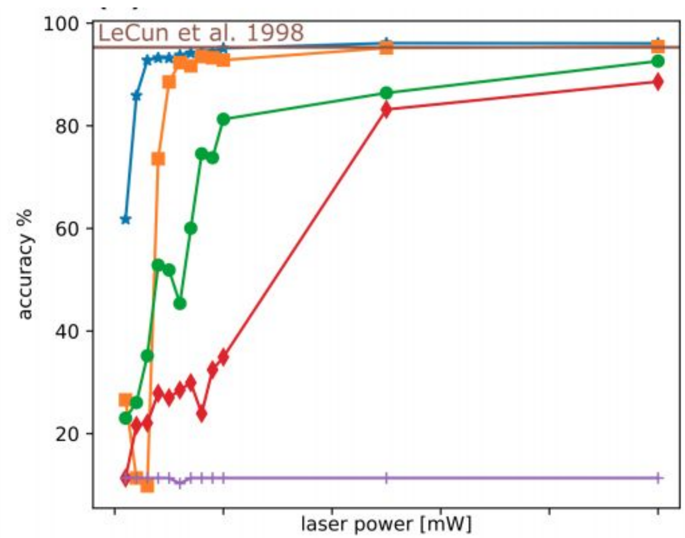
Name	Plot	Equation	Derivative (with respect to *x*)	Range
Identity		$f(x) = x$	$f'(x) = 1$	$(-\infty, \infty)$
Binary step		$f(x) = \begin{cases} 0 & \text{for } x < 0 \\ 1 & \text{for } x \geq 0 \end{cases}$	$f'(x) = \begin{cases} 0 & \text{for } x \neq 0 \\ ? & \text{for } x = 0 \end{cases}$	$\{0, 1\}$
Logistic (a.k.a. Soft step)		$f(x) = \frac{1}{1 + e^{-x}}$	$f'(x) = f(x)(1 - f(x))$	$(0, 1)$
TanH		$f(x) = \tanh(x) = \frac{2}{1 + e^{-2x}} - 1$	$f'(x) = 1 - f(x)^2$	$(-1, 1)$
ArcTan		$f(x) = \tan^{-1}(x)$	$f'(x) = \frac{1}{x^2 + 1}$	$\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$
Softsign [7][8]		$f(x) = \frac{x}{1 + \lvert x \rvert}$	$f'(x) = \frac{1}{(1 + \lvert x \rvert)^2}$	$(-1, 1)$
Inverse square root unit (ISRU)[9]		$f(x) = \frac{x}{\sqrt{1 + \alpha x^2}}$	$f'(x) = \left(\frac{1}{\sqrt{1 + \alpha x^2}}\right)^3$	$\left(-\frac{1}{\sqrt{\alpha}}, \frac{1}{\sqrt{\alpha}}\right)$
Rectified linear unit (ReLU)[10]		$f(x) = \begin{cases} 0 & \text{for } x < 0 \\ x & \text{for } x \geq 0 \end{cases}$	$f'(x) = \begin{cases} 0 & \text{for } x < 0 \\ 1 & \text{for } x \geq 0 \end{cases}$	$[0, \infty)$
Leaky rectified linear unit (Leaky ReLU)[11]		$f(x) = \begin{cases} 0.01x & \text{for } x < 0 \\ x & \text{for } x \geq 0 \end{cases}$	$f'(x) = \begin{cases} 0.01 & \text{for } x < 0 \\ 1 & \text{for } x \geq 0 \end{cases}$	$(-\infty, \infty)$

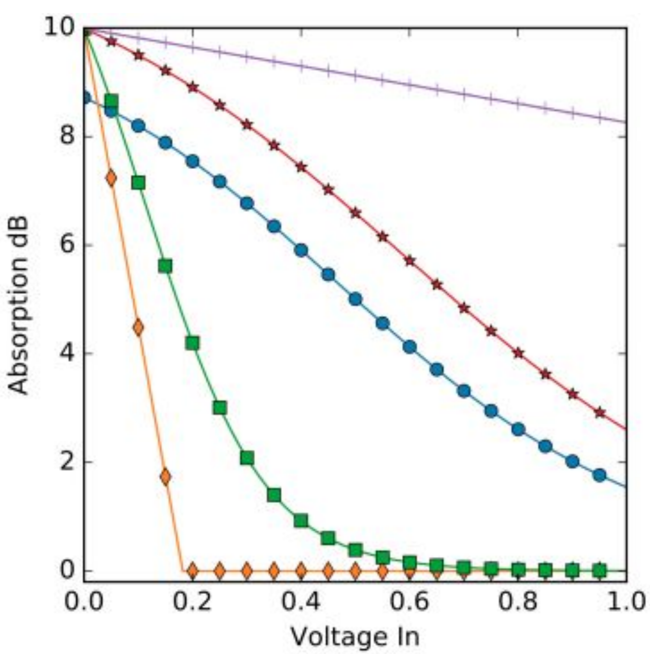
Results

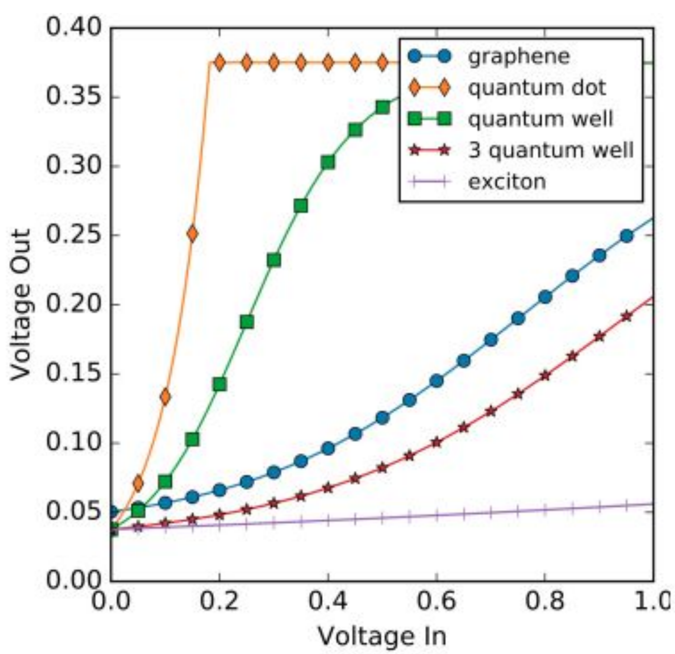
Design Space Explorations – Nonlinear Activation Functions

Explore and optimize nonlinear activation function (NAL) design

- Dynamic range of voltage is dependent on modulator length and laser power
- Modulator length determines how much light is absorbed







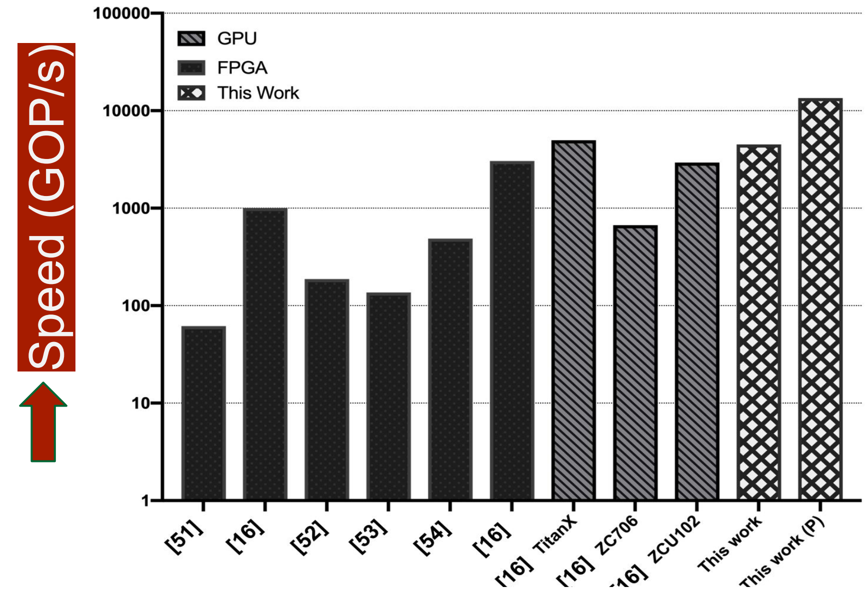
George, J. K., Mehrabian, A., Amin, R., Meng, J., De Lima, T. F., Tait, A. N., T, El-Ghazawi, & Sorger, V. J. (2019). Neuromorphic photonics with electro-absorption modulators. Optics express, 27(4), 5181-5191.

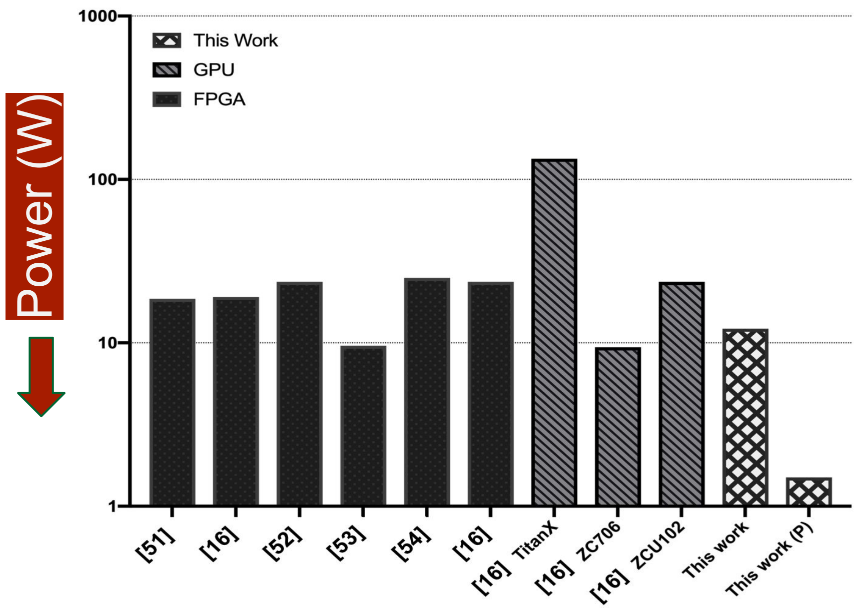
George J, Amin R, Mehrabian A, Khurgin J, El-Ghazawi T, Prucnal PR, Sorger VJ. Electrooptic nonlinear activation functions for vector matrix multiplications in optical neural networks. InSignal Processing in Photonic Communications 2018 Jul 2 (pp. SpW4G-3). Optical Society of America.

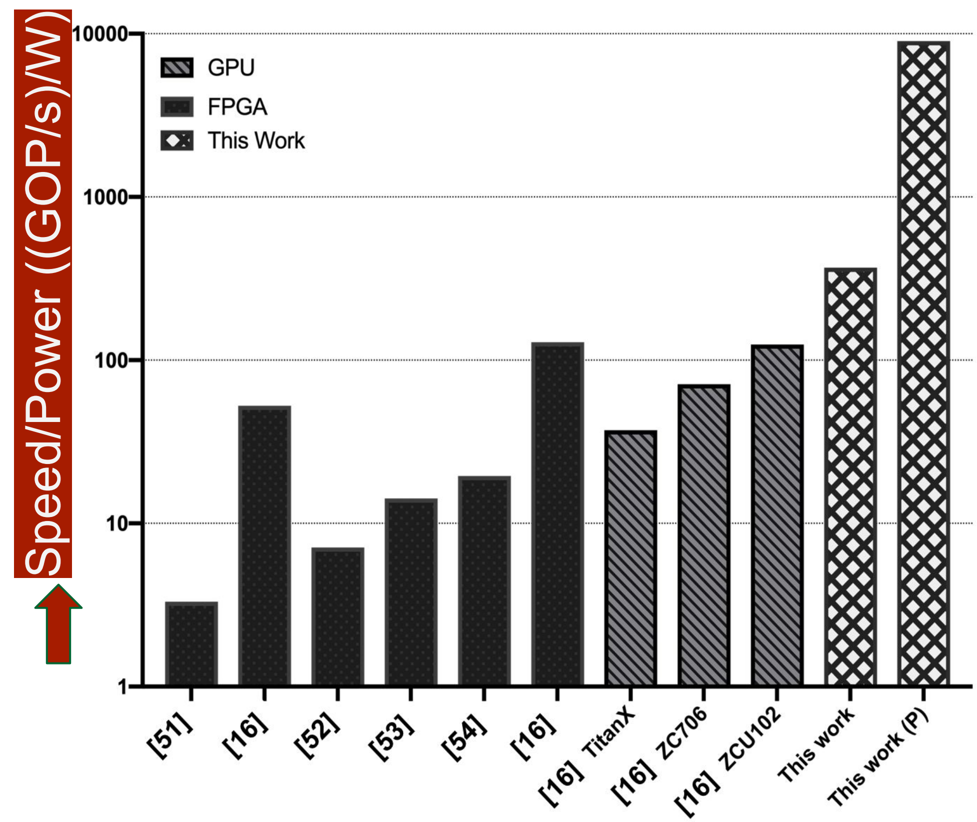
Results

Performance of the Proposed Winograd CNN Accelerator

- Benchmarked with **VGG16**
- **VGG16** has only convolutional layers



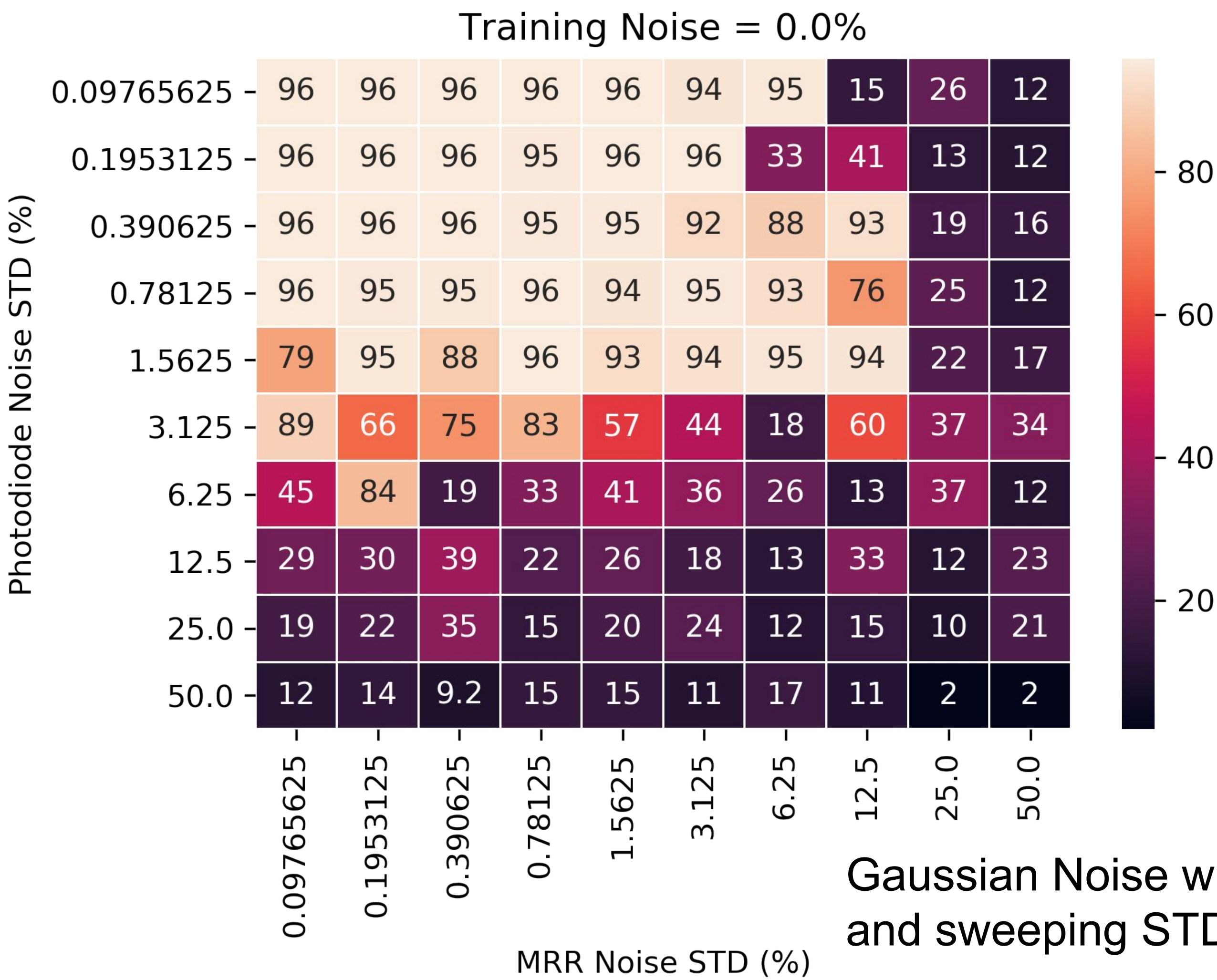




Results

Hardware-aware Software Design (The Effect of Noise)

- The effect of photodiode and MRR noise on prediction accuracy
- Noise only added during inference time (not training)

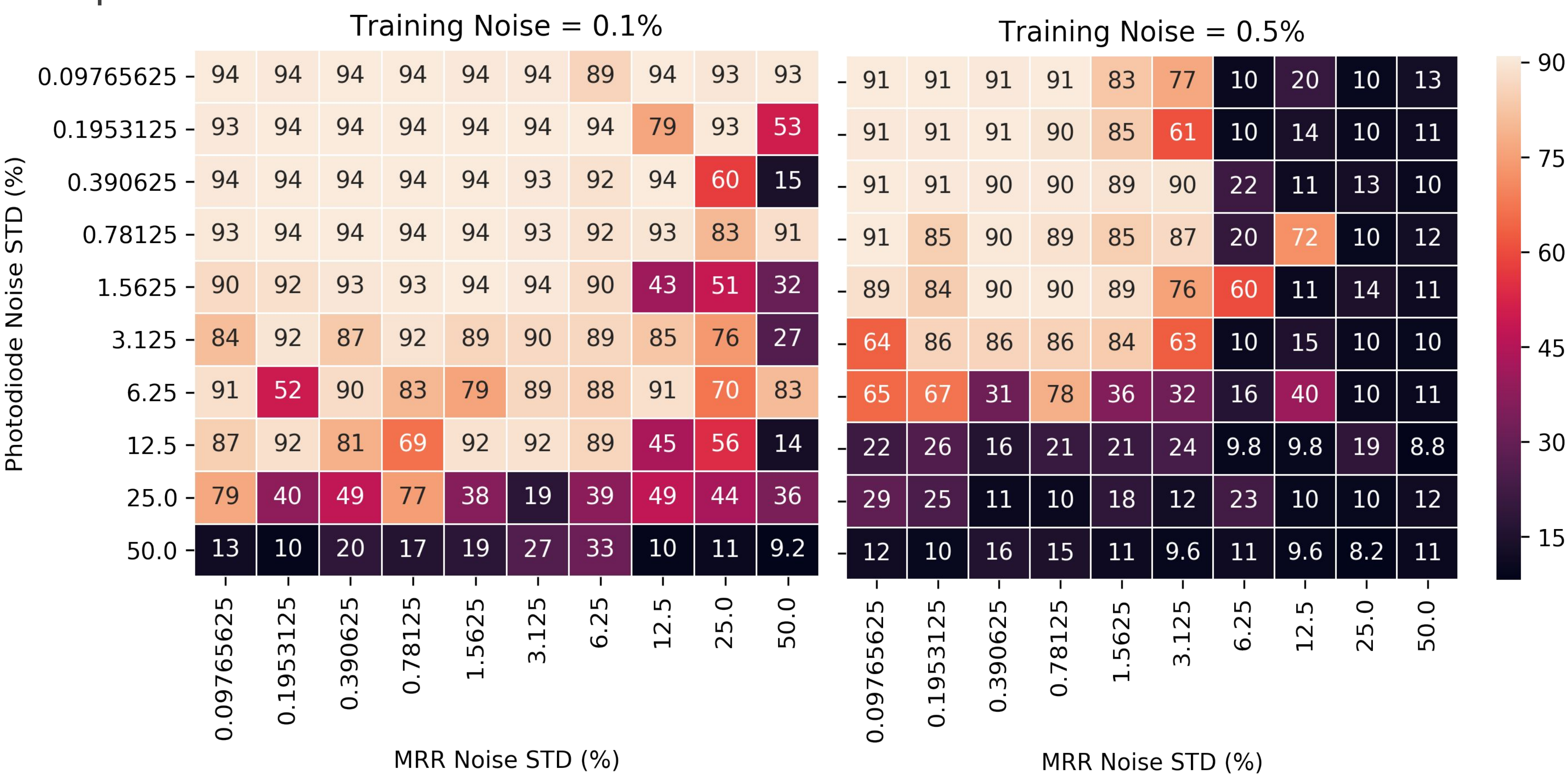


Gaussian Noise with **mean = 0** and sweeping STD

Results – Hardware-aware Software Design

The Effect of Noise

- The addition of slight amount of noise during the training can make the neural network more resilient against inference time noise
- Further addition of noise during training would deteriorate the inference performance



Gaussian Noise with **mean = 0** and sweeping STD

Approach
Envisioning and Optical Brain Simulator

Brain Simulators

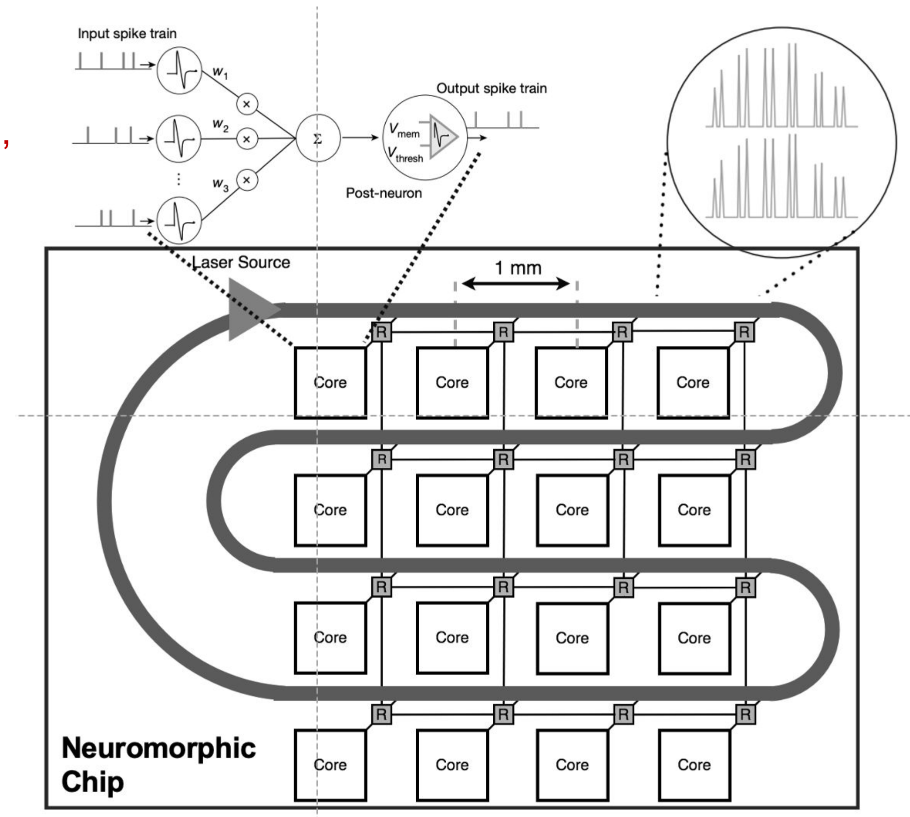
- Called “neuronal” networks,
  - To differentiate from “neural” networks
- Software applications that simulate a neuronal networks
  - As a whole -> full brain
  - In part -> Single functionality/region of the brain
- Rely on computational/mathematical models of neurons
  - Simple models -> less computational load, less accuracy
  - Complicated models -> more computational load, more accuracy
  - From simple arithmetic/logic to solving PDEs

- NEURON
- Genesis
- BRIAN
- NEST
- C2-simulator

Approach: Adaptive many-core brain simulator through   on a hybrid reconfigurable optical NoC.

Key Features

- Simple photonic processing cores
- Custom interconnect fabric
- Departure from conventional parallel machine principles
- Fault-tolerant Design



Approach

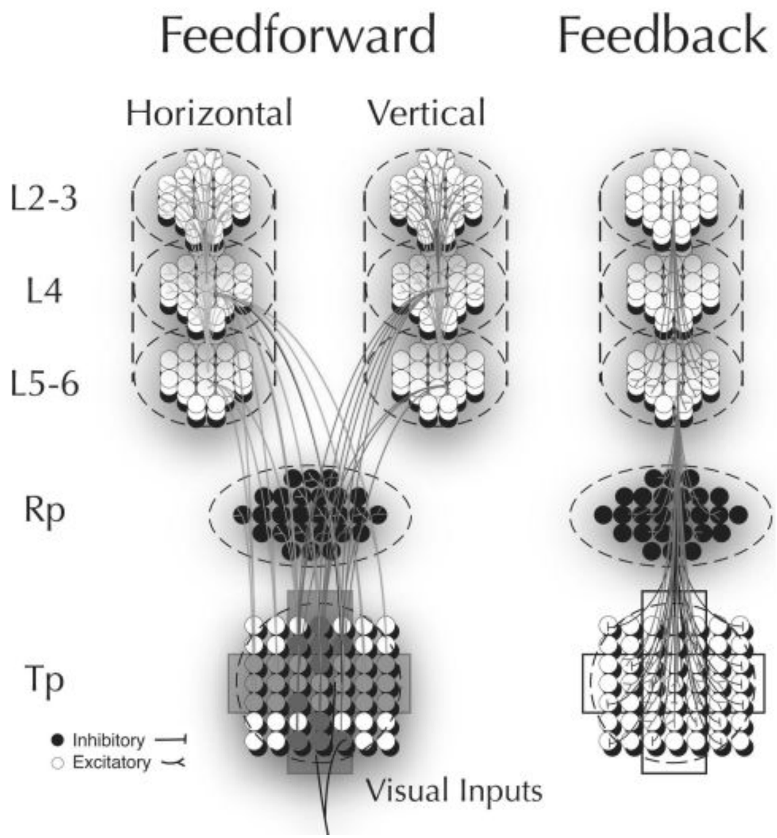
Simulation of the Hill-Tononi Model

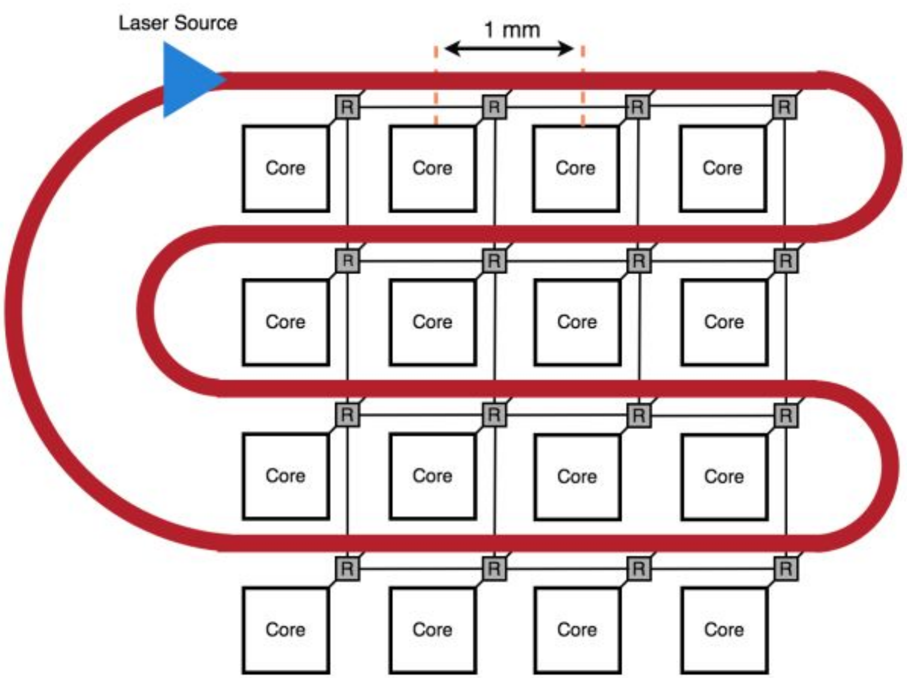
GW

- Models the brains transition from wakefulness to sleep
- The model consists of
  - two visual areas and
  - associated thalamic and reticular thalamic nuclei
- During wakefulness, cortical neurons fire at irregular intervals
- With the onset of slow-wave sleep,
  - virtually all cortical neurons undergo a slow oscillation (<1 Hz) in their membrane potential
  - followed by neurons fire at rates that are even higher than in quiet wakefulness lasting several hundred milliseconds

Approach
Simulation of the Hill-Tononi Model

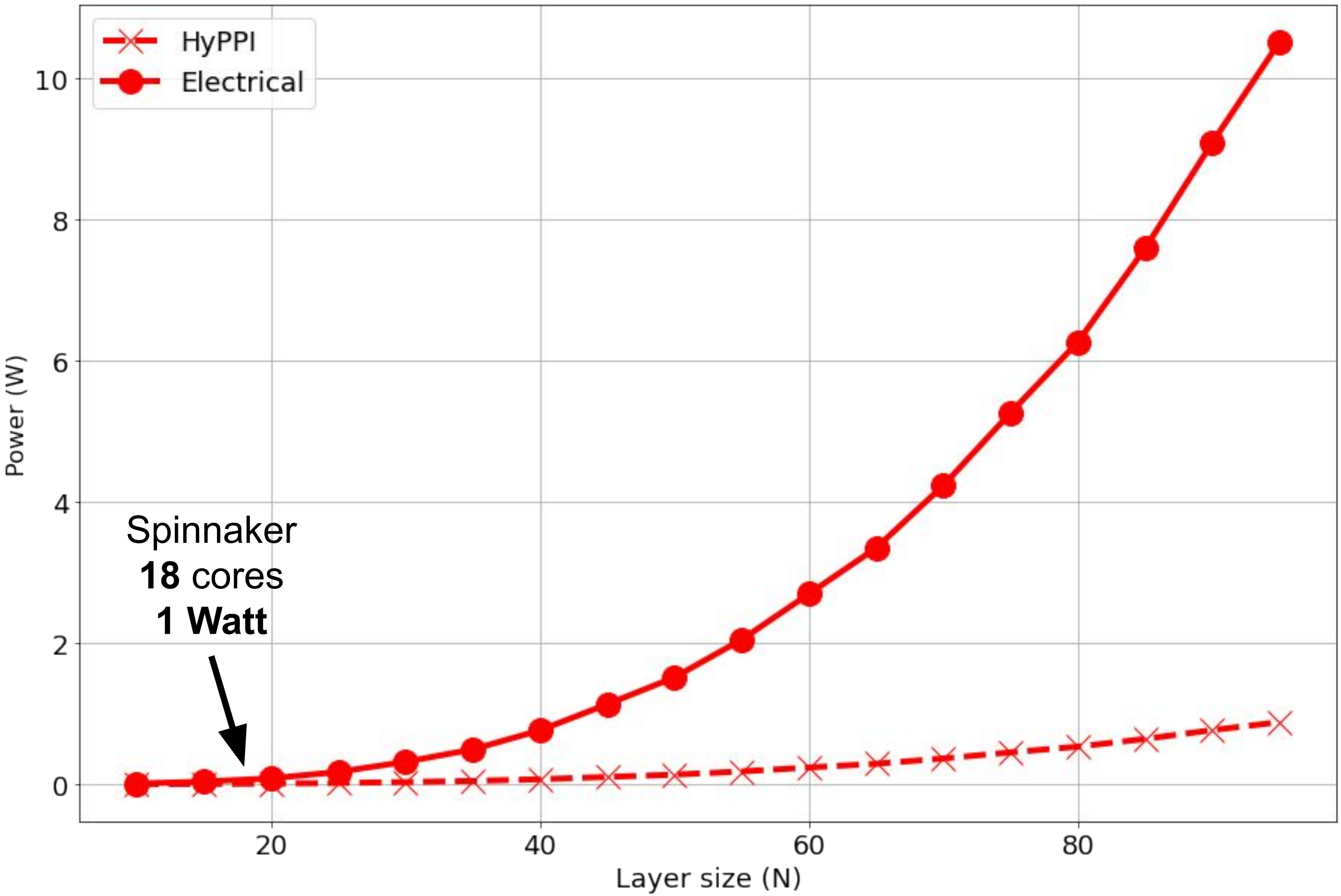
- Each neuron is processed on a single core
- Inter-layer neurons connect through HyPPI





Result

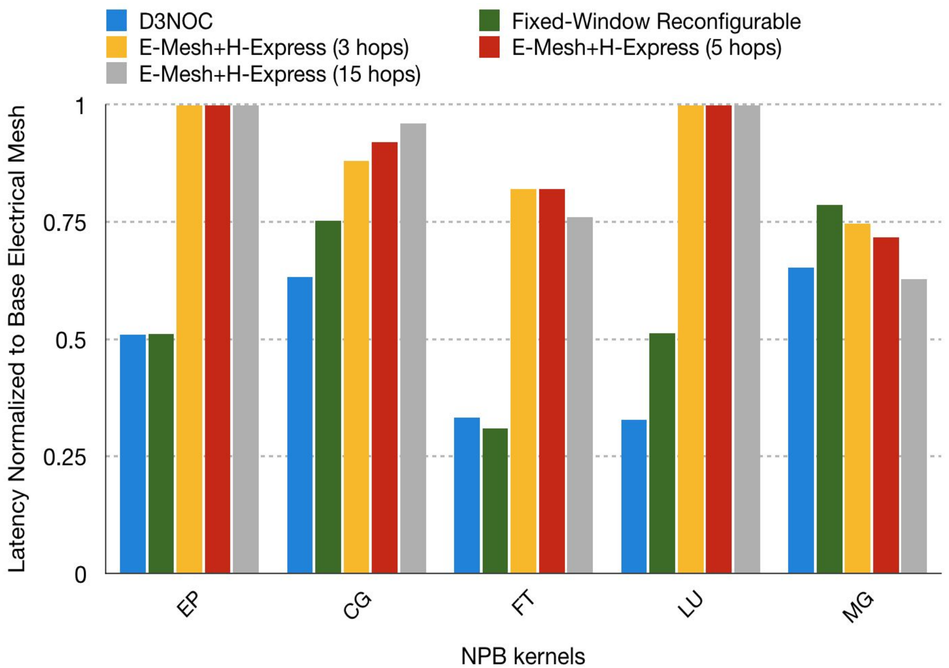
Power simulations (HyPPI vs Electrical)

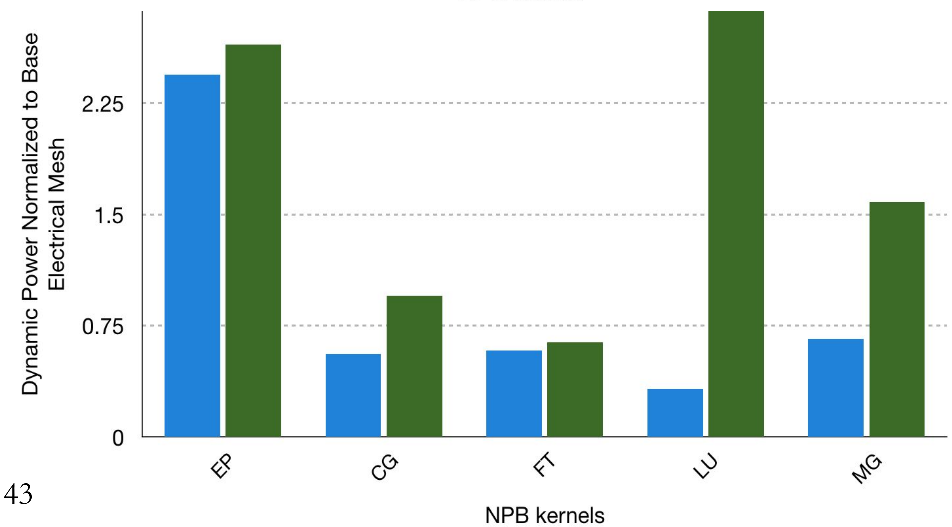


Results

Further Enhancements using an Adaptive HyPPI NoC

- An adaptive dynamic reconfigurable NoC design
- Adaptation based on changes in the NoC traffic are considered
- We monitor and predict traffic, then
- augment the topology with an express optical bus
- Up to **67%** saving in latency
- Up to **69%** saving in power





Mehrabian A, Sun S, Narayana VK, Anderson J, Peng J, Sorger V, El-Ghazawi T. D3NoC: a dynamic data-driven hybrid photonic plasmonic NoC. InProceedings of the 15th ACM International Conference on Computing Frontiers 2018 May 8 (pp. 220-223).

Conclusion

- As transistors reach atomic scale the classical technological driver that has underpinned Moore's Law for 50 years is failing and is anticipated to flatten by 2025.
- New models of computation such as neuromorphic is needed to solve problems that are not well addressed by digital computing
- We can exploit the synergy between neuromorphic computing and photonic computing for the next generation of our computers
- In order to realize full-scale photonic neural networks we need to take advantage of any conceptual (mathematical transforms) and/or physical tool (memristors, eam) in our arsenal
- Photonic neural networks can potentially push the limits of electronic implementations by orders of magnitude (limited by shortcoming of electronics)
- The problem of brain simulation is communication-bound and novel optical interconnects have the potential to improve the shortcomings of the electrical interconnects

Peer-reviewed publications and patent

1. "Mehrabian, Armin; Miscuglio, Mario; Alkabani, Yousra; Sorger, Volker J; El-Ghazawi, Tarek; ",A Winograd-based Integrated Photonics Accelerator for Convolutional Neural Networks,IEEE Journal of Selected Topics in Quantum Electronics,26,1,1-12,2019,IEEE
2. “Mehrabian, Armin, Volker J. Sorger, and Tarek El-Ghazawi. "A Design Methodology for Post-Moore’s Law Accelerators: The Case of a Photonic Neuromorphic Processor." 2020 IEEE 31st International Conference on Application-specific Systems, Architectures and Processors (ASAP). IEEE, 2020. APA
3. "Sun, Shuai; Narayana, Vikram K; Mehrabian, Armin; El-Ghazawi, Tarek; Sorger, Volker J; ",A universal multi-hierarchy figure-of-merit for on-chip computing and communications,arXiv preprint arXiv:1612.02486,,,,2016,
4. "Narayana, Vikram K; Sun, Shuai; Mehrabian, Armin; Sorger, Volker J; El-Ghazawi, Tarek; ",HyPPI NoC: Bringing hybrid plasmonics to an opto-electronic network-on-chip,2017 46th International Conference on Parallel Processing (ICPP),,,131-140,2017,IEEE
5. "Mehrabian, Armin; Sun, Shuai; Narayana, Vikram K; Sorger, Volker J; El-Ghazawi, Tarek; ",D3noc: Dynamic data-driven network on chip in photonic electronic hybrids,arXiv preprint arXiv:1708.06721,,,,2017,
6. "Sun, Shuai; Narayana, Vikram K; Mehrabian, Armin; Zhang, Ruoyu; El-Ghazawi, Tarek; Sorger, Volker J; ",Holistic Performance-Cost Metric for Post Moore Era,Frontiers in Optics,,,JTu2A. 24,2017,Optical Society of America
7. "George, Jonathan; Amin, Rubab; Mehrabian, Armin; Khurgin, Jacob; El-Ghazawi, Tarek; Prucnal, Paul R; Sorger, Volker J; ",Electrooptic nonlinear activation functions for vector matrix multiplications in optical neural networks,Signal Processing in Photonic Communications,,,SpW4G. 3,2018,Optical Society of America
8. "Mehrabian, Armin; Al-Kabani, Yousra; Sorger, Volker J; El-Ghazawi, Tarek; ",PCNNA: a photonic convolutional neural network accelerator,2018 31st IEEE International System-on-Chip Conference (SOCC),,,169-173,2018,IEEE
9. "Mehrabian, Armin; Sun, Shuai; Narayana, Vikram K; Anderson, Jeff; Peng, Jiaxin; Sorger, Volker; El-Ghazawi, Tarek; ",D3NoC: a dynamic data-driven hybrid photonic plasmonic NoC,Proceedings of the 15th ACM International Conference on Computing Frontiers,,,220-223,2018,
10. "George, Jonathan K; Mehrabian, Armin; Amin, Rubab; Meng, Jiawei; De Lima, Thomas Ferreira; Tait, Alexander N; Shastri, Bhavin J; El-Ghazawi, Tarek; Prucnal, Paul R; Sorger, Volker J; ",Neuromorphic photonics with electro-absorption modulators,Optics express,27,4,5181-5191,2019,Optical Society of America
11. "George, Jonathan; Mehrabian, Amin; Amin, Rubab; El-Ghazawi, Tarek; Prucnal, Paul K; Sorger, Volker J; ",Photonic Neural Network Nonlinear Activation Functions by Electrooptic Absorption Modulators,Frontiers in Optics,,,JW3A. 123,2018,Optical Society of America
12. "Miscuglio, Mario; Mehrabian, Armin; Hu, Zibo; Azzam, Shaimaa I; George, Jonathan; Kildishev, Alexander V; Pelton, Matthew; Sorger, Volker J; ",All-optical nonlinear activation function for photonic neural networks,Optical Materials Express,8,12,3851-3863,2018,Optical Society of America
13. "George, Jonathan; Mehrabian, Armin; Amin, Rubab; Prucnal, Paul R; El-Ghazawi, Tarek; Sorger, Volker J; ",Neural network activation functions with electro-optic absorption modulators,2018 IEEE International Conference on Rebooting Computing (ICRC),,,1-5,2018,IEEE
14. "Anderson, Jeff; Mehrabian, Armin; Peng, Jiaxin; El-Ghazawi, Tarek A; ",Extreme Heterogeneity in Deep Learning Architectures.,,,,,2019,
15. "George, Jonathan K; Mehrabian, Amin; Amin, Rubab; El-Ghazawi, Tarek; Prucnal, Paul R; Sorger, Volker J; ",Photonic Neuromporphic Computing with Electrooptic Nonlinear Activation,2018 Photonics in Switching and Computing (PSC),,,1-3,2018,IEEE
16. "George, JK; Mehrabian, A; ",Photonic Neuromorphic Processors and Convolutional Neural Network Accelerator,Cognitive Computing,,,,2018,
17. Miscuglio, Mario, et al. "Artificial synapse with mnemonic functionality using GSST-based photonic integrated memory." 2020 International Applied Computational Electromagnetics Society Symposium (ACES). IEEE, 2020.
18. "Miscuglio, Mario; Yesiliurt, Omer; Meng, Jiawei; Prokopeva, Ludmila J; Zhang, Yifei; Mehrabian, Armin; Hu, Juejun; Kildishev, Alexander V; Sorger, Volker J; ",Intelligent Computing with Photonic Memories,2020 Optical Fiber Communications Conference and Exhibition (OFC),,,1-3,2020,IEEE
19. **Patent:** "Mehrabian, Armin; Sorger, Volker J; El-Ghazawi, Tarek; Miscuglio, Mario; ",Optical convolutional neural network accelerator,,,,,2020,"US Patent App. 16/507,854”

Backup Slides

Related Work

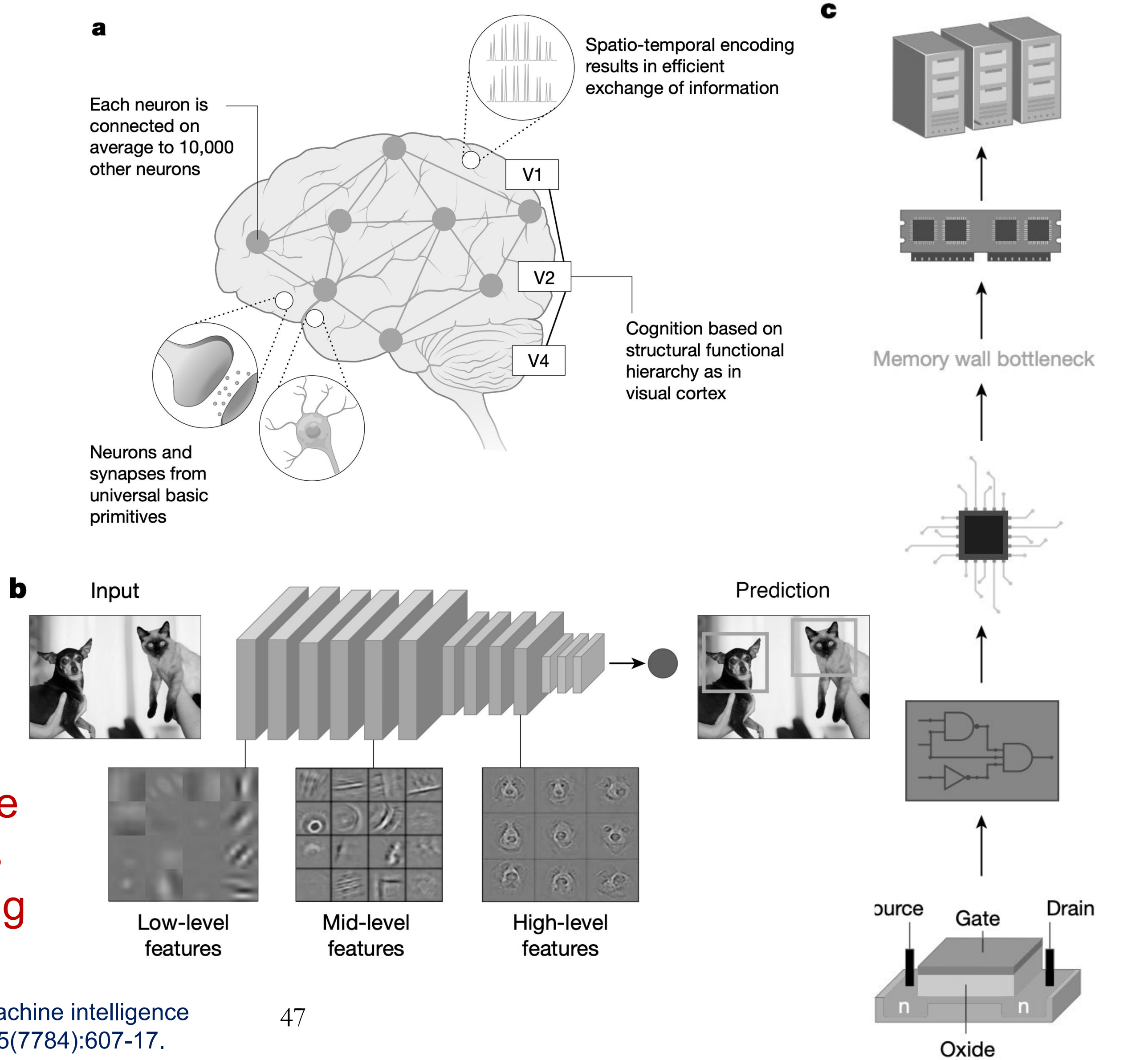
ANNs vs. Neuromorphic

a. The intertwined network of neurons and synapses with temporal spike processing enables rapid and efficient flow of information between different areas

b. What we know as ANN today aim to mimic functional and algorithmic methods of brain while remain mostly **hardware-agnostic**

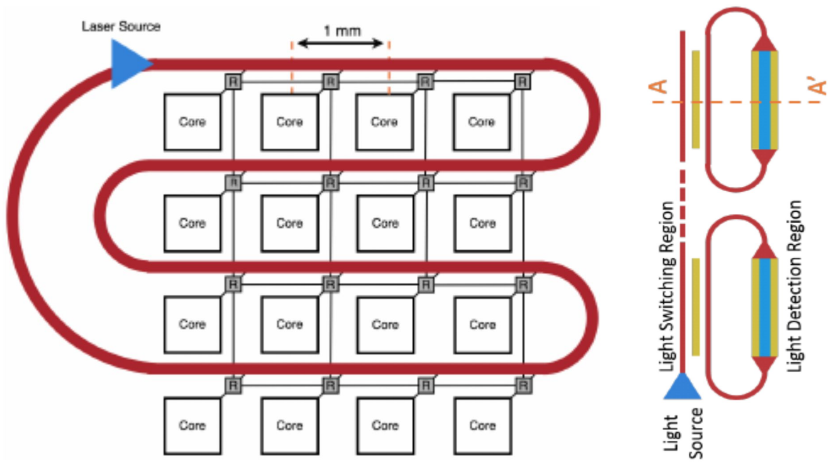
c. Our digital Von-Neumann computers are the mainstream hardware platform to execute ANNs

Artificial Neural Networks are conceptual vehicles towards fully neuromorphic computing



Roy K, Jaiswal A, Panda P. Towards spike-based machine intelligence with neuromorphic computing. Nature. 2019 Nov;575(7784):607-17.

Envisioning an Optical Brain Simulator



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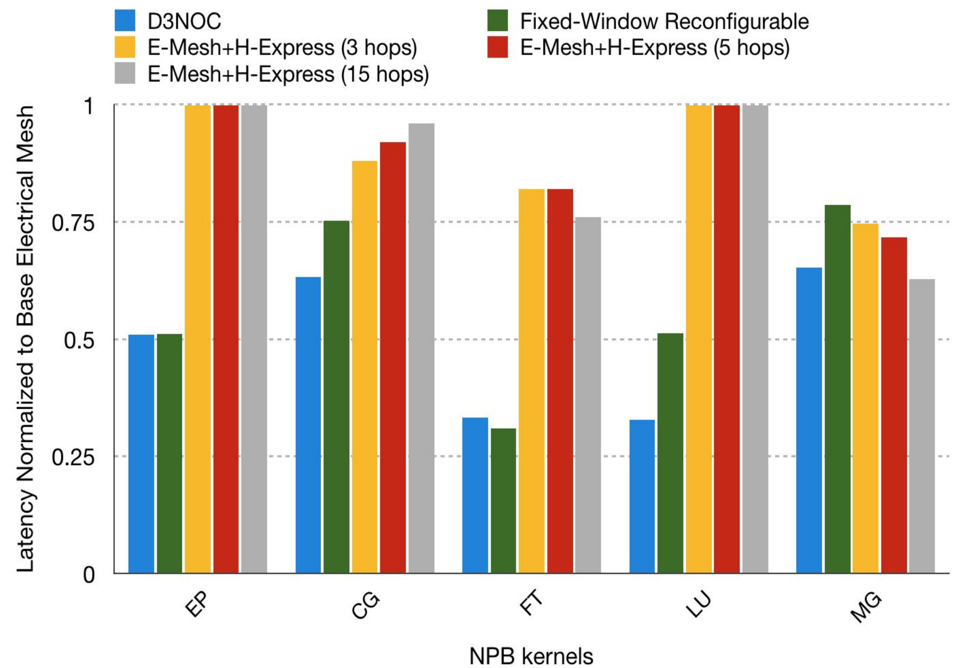

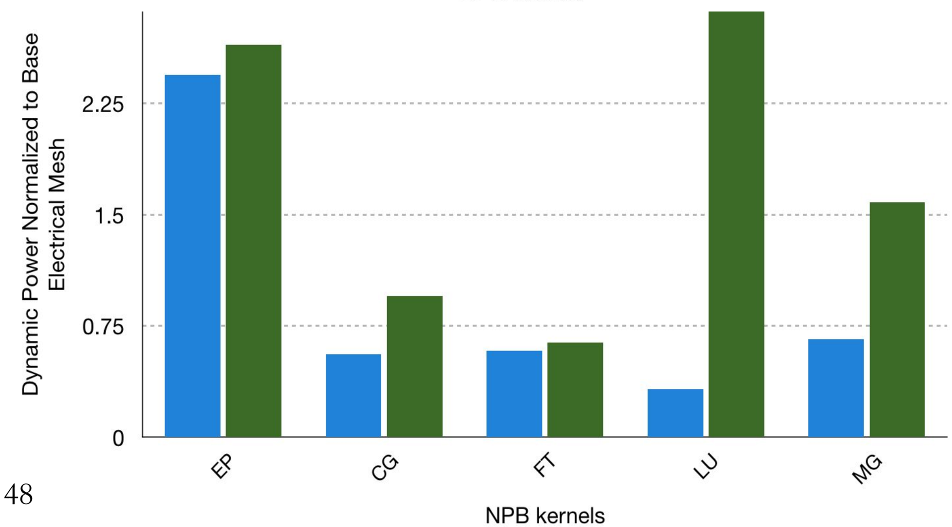

Mehrabian A, Sun S, Narayana VK, Anderson J, Peng J, Sorger V, El-Ghazawi T. D3NoC: a dynamic data-driven hybrid photonic plasmonic NoC. InProceedings of the 15th ACM International Conference on Computing Frontiers 2018 May 8 (pp. 220-223).

Envisioning an Optical Brain Simulator

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- ▸ Called "neuronal" networks,
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- ▸ Software applications that simulate a neuronal networks
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- NEURON
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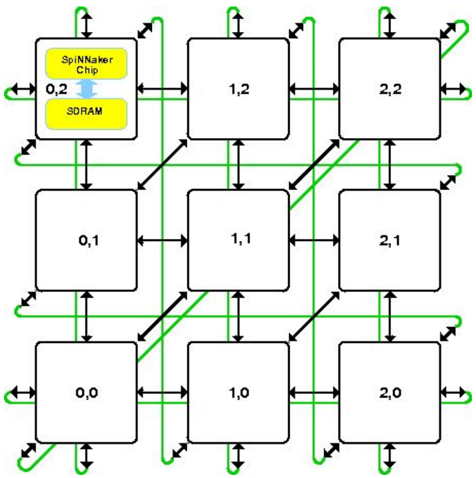
Approach: Adaptive many-core brain simulator through on a hybrid reconfigurable optical NoC.

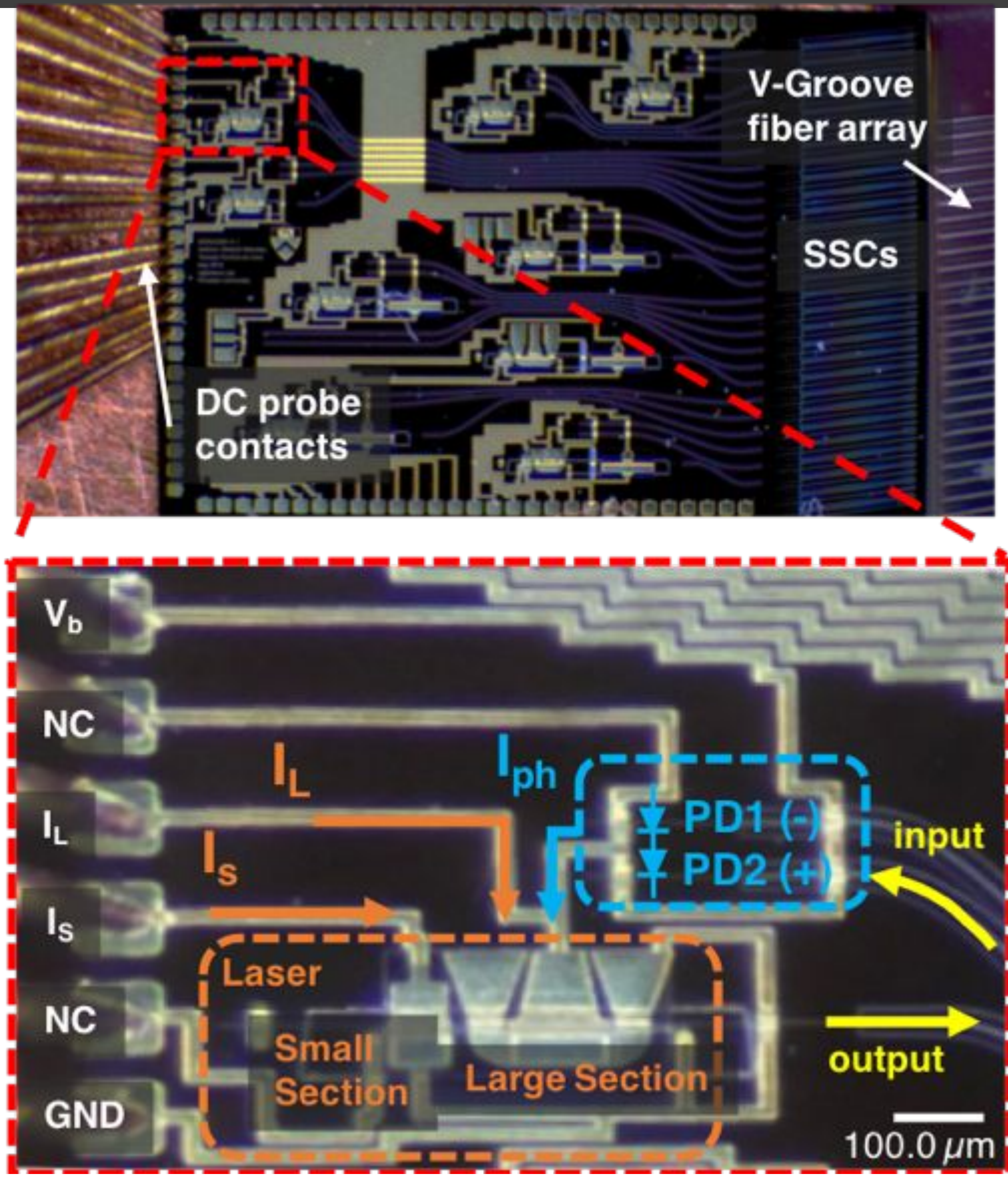
Key Features

- Simple photonic processing cores
- Custom interconnect fabric
- Departure from conventional parallel machine principles
- Fault-tolerant Design

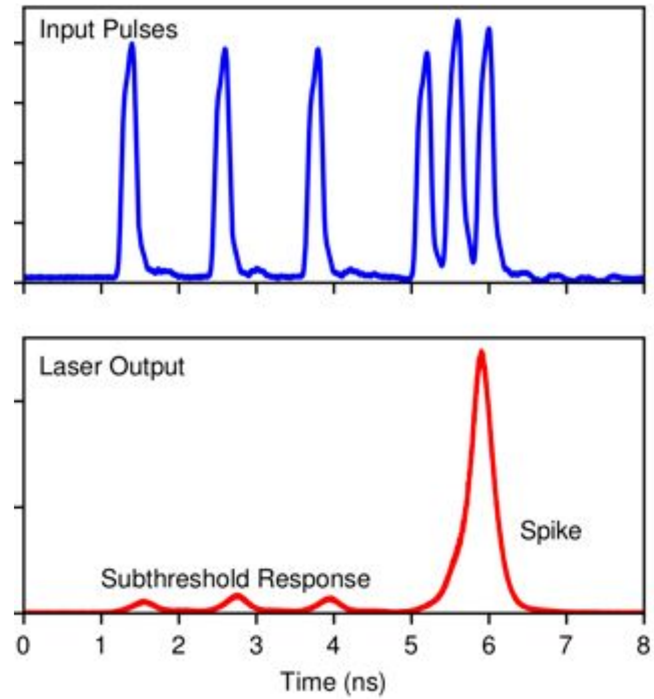
System Architecture



III-V Photonic Neuron Implementation



Peng et al. *IEEE JSTQE* (Submitted)



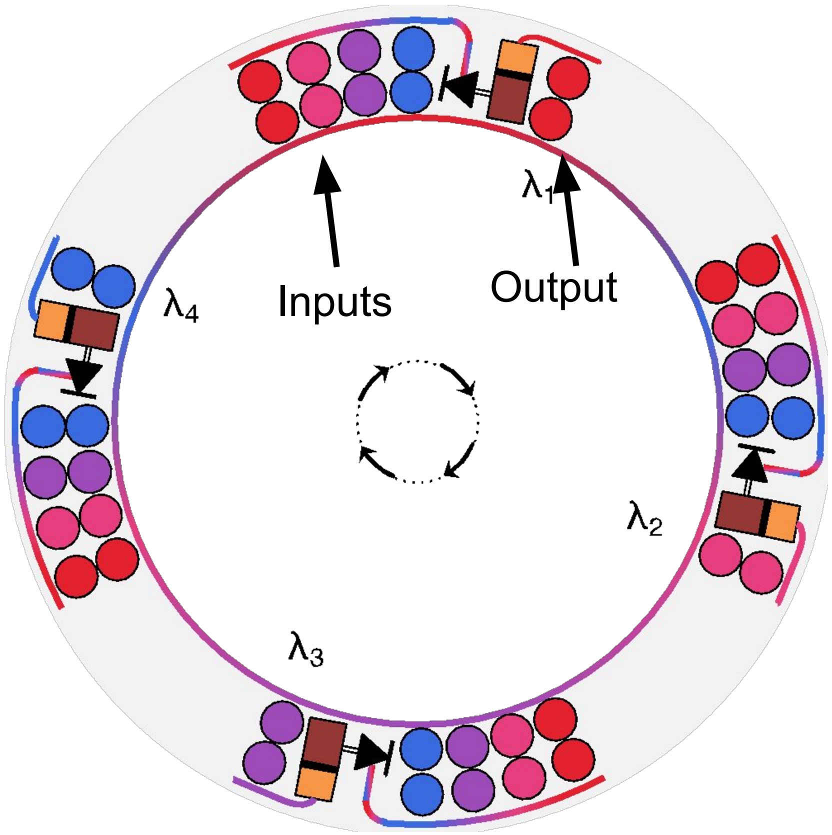
- ❑ Experimentally confirmed that an integrated laser (left) can behave like a biological spiking neuron at ultrafast time scales
- ❑ Currently testing a laser neural network (top left) that can perform spatio-temporal processing tasks

Synergy

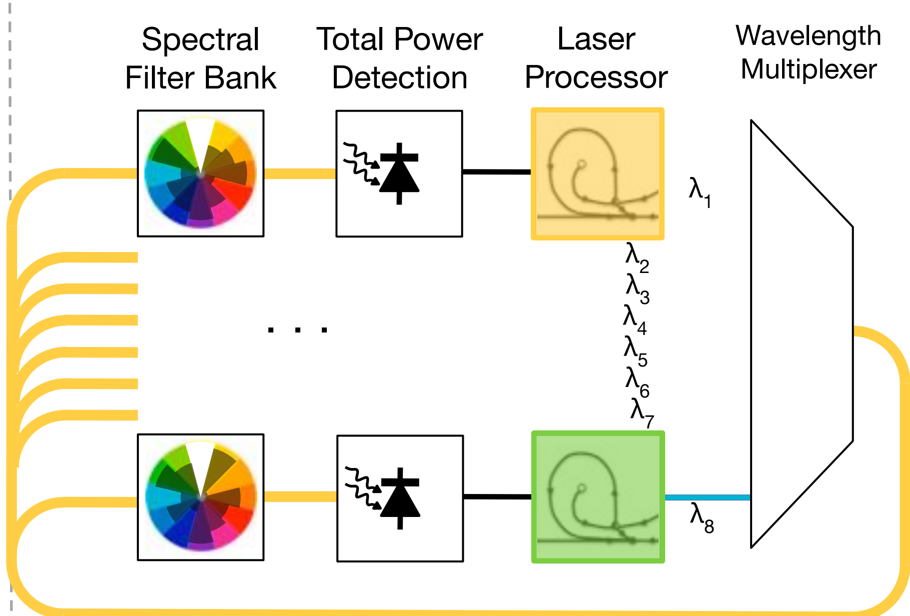
- CNNs have sparse connection
- Sparse connections -> Less MRRs (hardware)
- From functional point of view CNNs prefer smaller kernels, 3x3, 5x5 etc.
- Winograd convolution reduces number of multiplications in a convolution operation
- Winograd convolution performs better with smaller kernel sizes
- Different kernels in a CNN layer are independent
- All kernels operate on the same input
- WDM can provide 10~100 fold parallelization

Small Systems

GW



Broadcast-and-Weight Network

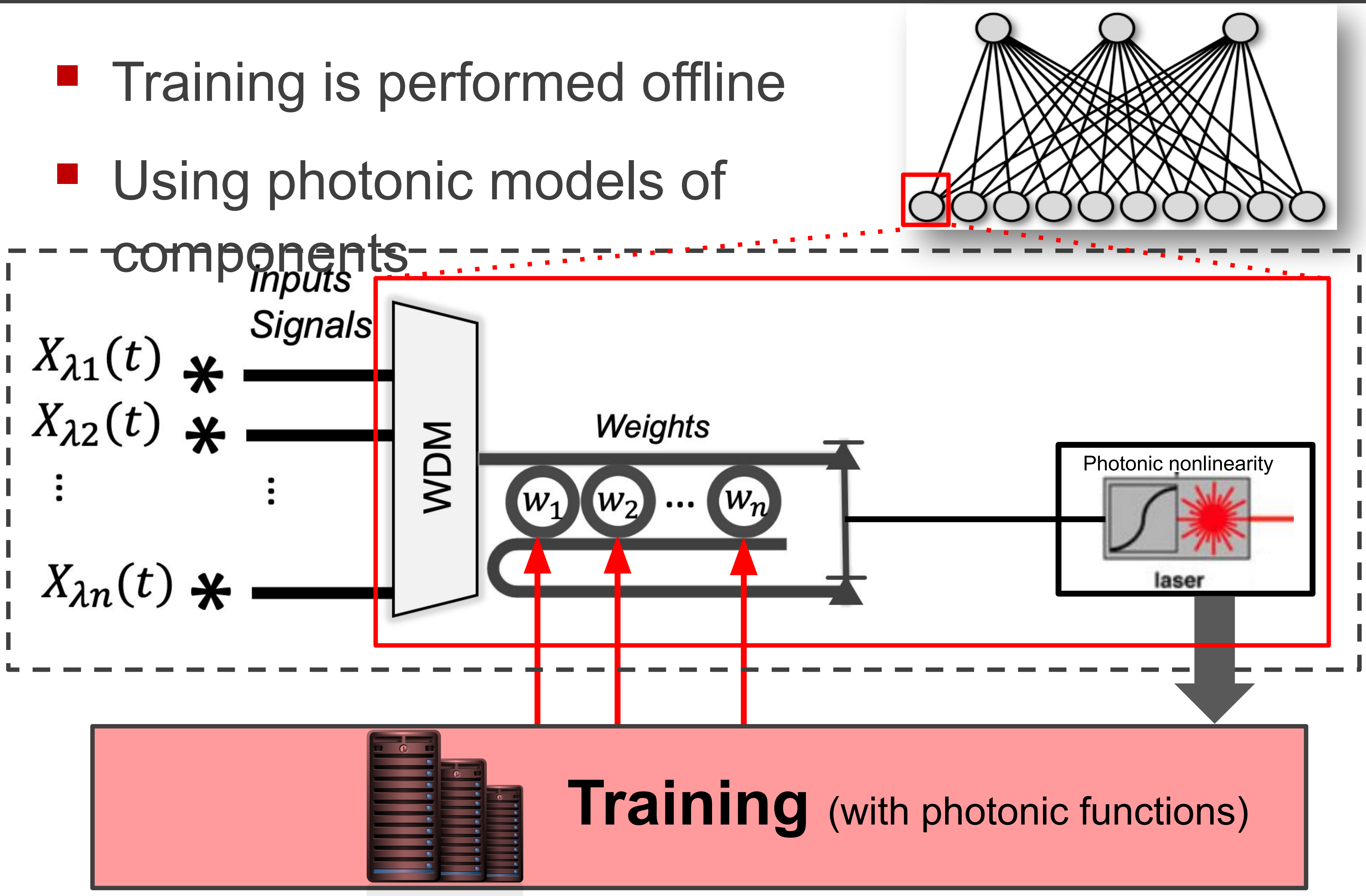


- N^2-connection interconnect in a **single** N-channel waveguide
- No switching, No routing
- **Bandwidth** <u>not</u> limited by interconnects

Tait et al. *J. Lightw. Technol.* **32** (2014)
Prucnal & Shastri *Neuromorphic Photonics* CRC Press (2017)

Photonic ANNs

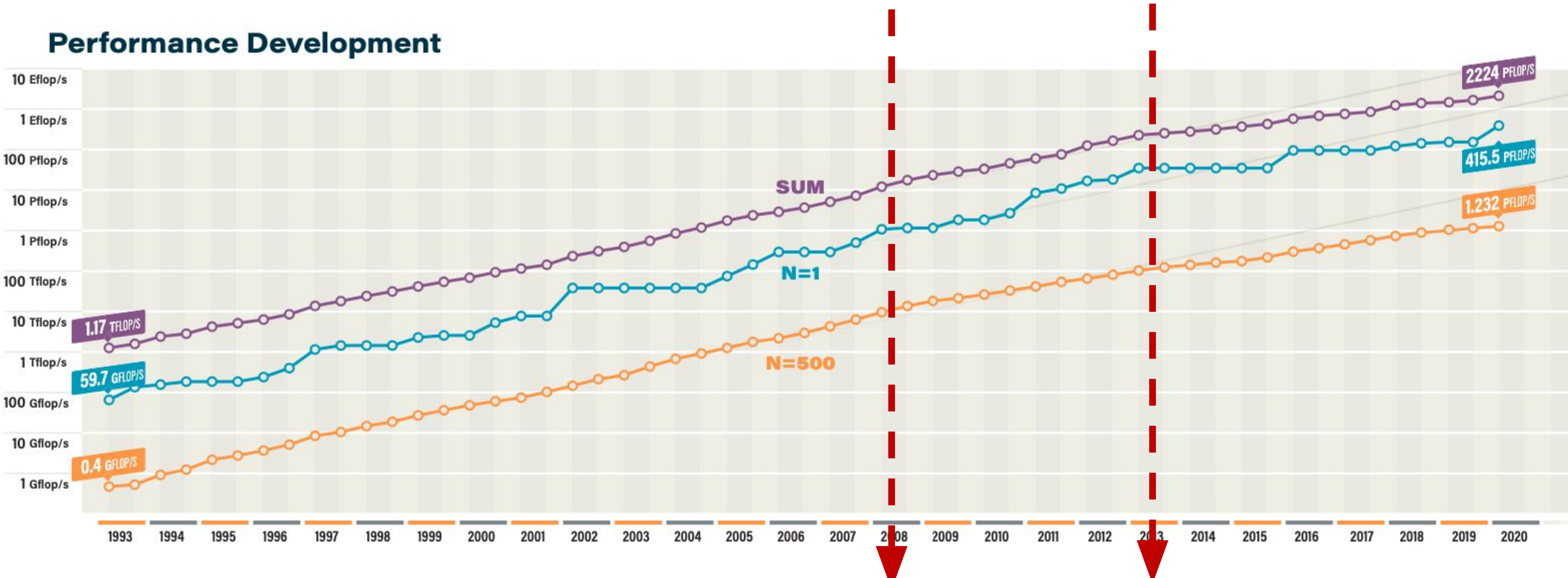
- Training is performed offline
- Using photonic models of components



Background and Motivation

Top 500 Supercomputer Performance

- We are on track to exa-scale systems – Fugaku
- High peak performance ~ **415.5 PetaFLOPS (HPL)**
- Power efficiency ~**14.66GFLOPS/Watt (#9 on Green500)**



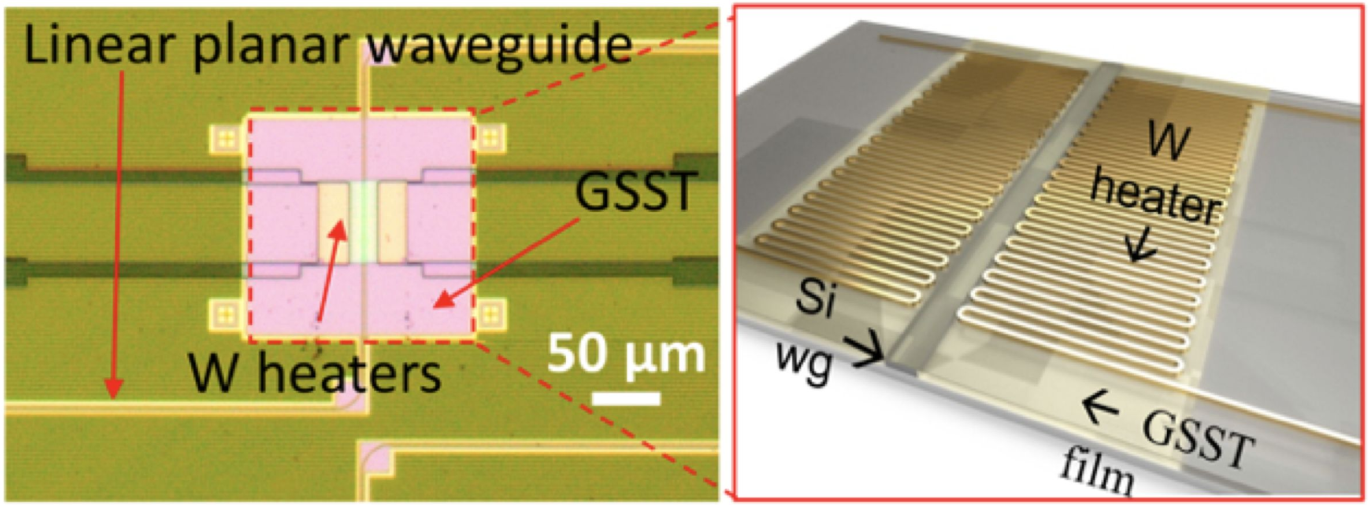
2008
Dennard's Scaling started to slow down

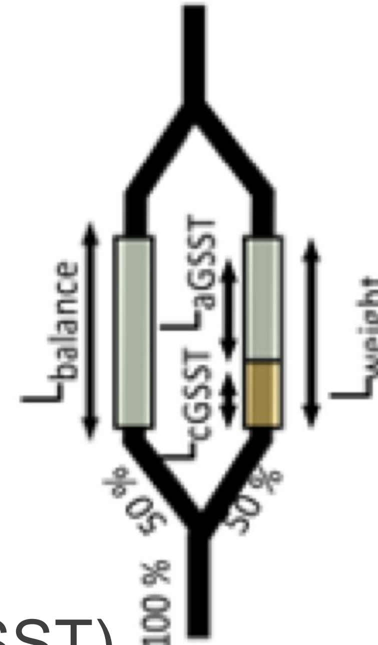
2013
We are on a less steep projection

http://Top500.org – June 2020

Accomplishment

GSST photonic integrated memory





- Hybrid silicon-Ge–Sb–Se–Te [$Ge_2Sb_2Se_4Te_1$] (GSST) electro-refractive modulator
- Shows coherent quantized response as function of the portion of the phase change (MHz programming Speed)

Miscuglio, M., Meng, J., Yesiliurt, O., Zhang, Y., Prokopeva, L. J., Mehrabian, A., ... & Sorger, V. J. (2020, July). Artificial synapse with mnemonic functionality using GSST-based photonic integrated memory. In 2020 International Applied Computational Electromagnetics Society Symposium (ACES) (pp. 1-3). IEEE. Chicago

To address the computation – communication disparity,

1. Need more efficient while communication/computation mitigating (or maintaining) the ratio
   - Nanophotonics (aJ/MAC)
   - Human brain (10s of Watts) vs Supercomputers (10s of Mwatts)
2. Bring data and computation close
   - Neuromorphic Computing
   - In-memory processing
3. With the explosion of data more and more people are looking at AI and data analytics
   - Demand for more specialized Hardware

Kendall, Jack D., and Suhas Kumar. "The building blocks of a brain-inspired computer." *Applied Physics Reviews* 7.1 (2020):

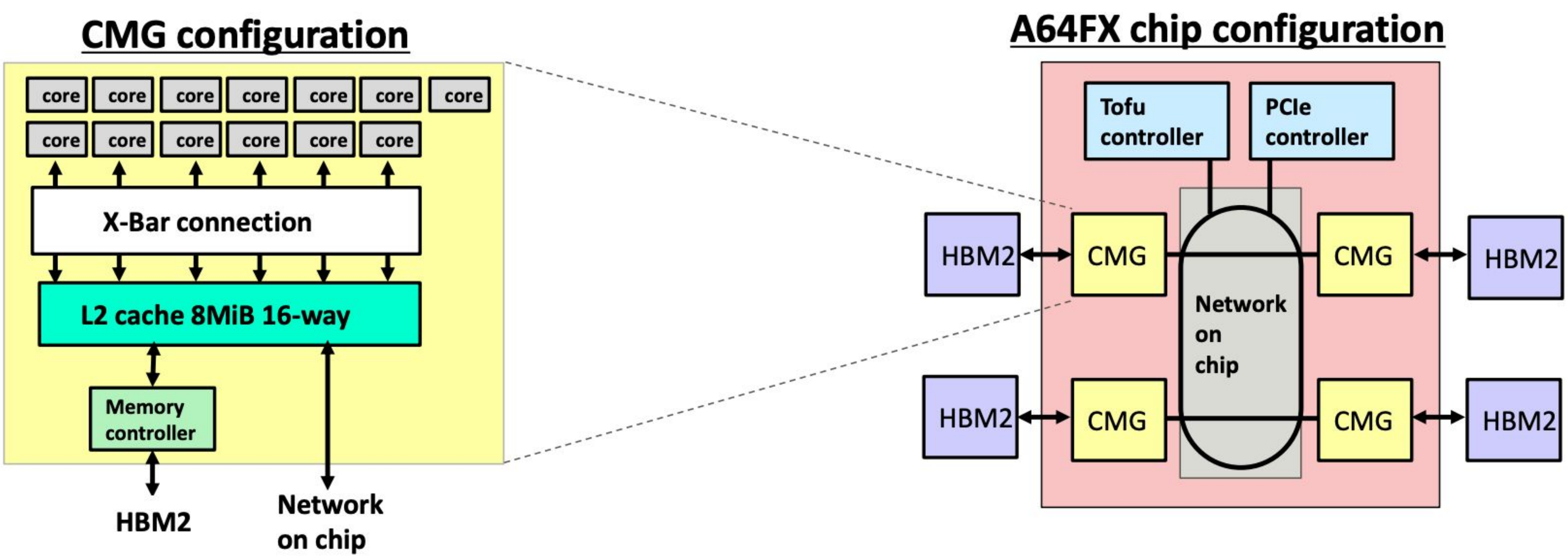
Top500 Nov19'

1. In the latest TOP500 rankings 56% of the additional flops are result of NVIDIA Tesla GPUs
2. Most of those additional flops came from top 3: Summit, Sierra, and the AI Bridging Cloud Infrastructure (ABCI).
3. 95% of the Summit's peak performance (187.7 petaflops) is derived from the system's 27,686 GPUs.

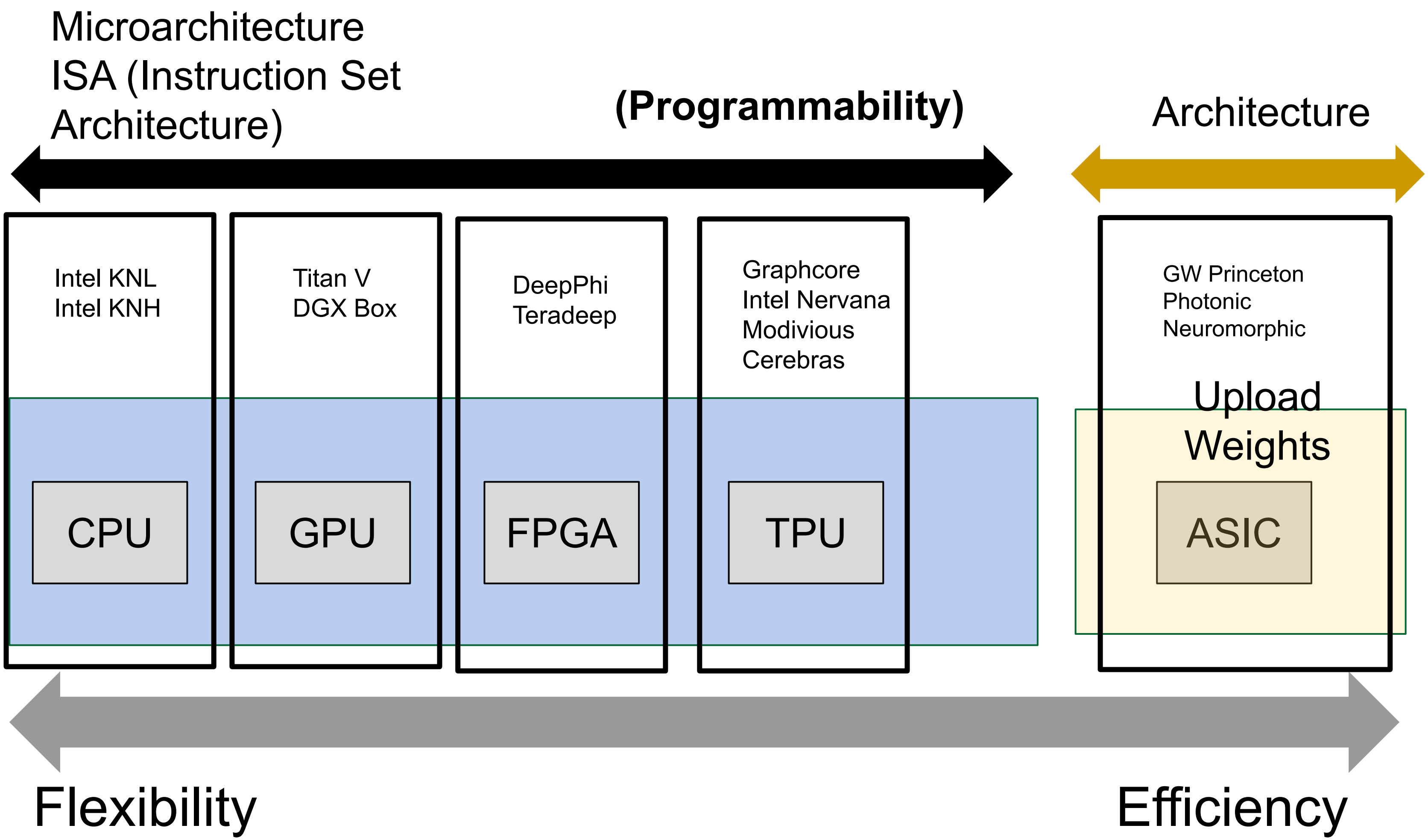
Fugaku the top supercomputer Jun20 (A64FX CPU or GPU?)

GW

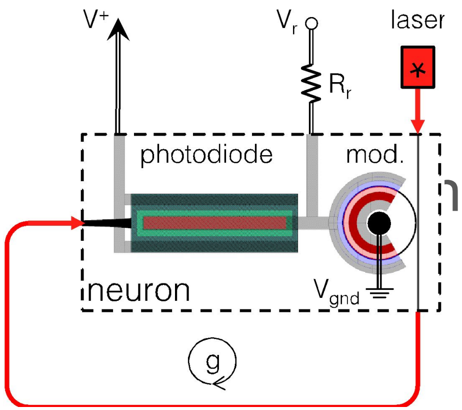
- While the cores in A64X chip are CPUs
- The architecture closely resembles GPUs
  - Many-core architecture
  - High vector performance due to wide vector lanes
  - Same memory system as NVIDIA Volta



- General Purpose vs Task Specific



MAC Efficiency and Scaling



n feedback loop:

$$g \equiv \frac{dP_{out}}{dP_{in}} = \frac{dP_{out}}{dv_{in}} \cdot \frac{dv_{in}}{di_{PD}} \cdot \frac{di_{PD}}{dP_{in}}$$

$$g = P_{pump} \left.\frac{dT}{dV}\right|_{max} R_b R_{PD}$$

$$g = P_{pump} \frac{\pi}{2V_\pi} R_b R_{PD}$$

$$\boxed{P_{pump} \geq \frac{2V_\pi}{\pi} \cdot \frac{1}{R_{PD}R_b} \cdot N_{FO}}$$

$$\boxed{\#MAC/s/neuron = N_{FI} \cdot f_{3dB},} \quad f_{3dB} = (2\pi R_b C_{mod})^{-1}.$$

$$\boxed{E_{\mathrm{MAC}} \geq \underbrace{N_{\mathrm{FO}}}_{\text{fan-out}} \cdot \underbrace{\frac{1}{\eta_{\mathrm{pp}}\eta_L\eta_{\mathrm{r}}}}_{\text{quantum efficiency}} \cdot \underbrace{\frac{h\nu}{N_{\mathrm{FI}}}}_{\text{photon energy/MAC}} \cdot \frac{1}{e} \underbrace{[2\pi V_s(C_{\mathrm{mod}} + C_{\mathrm{PD}})]}_{\text{switching charge for gain cascadability}}}$$

MACs/Second:

C_{total} = 47 fF
$V_s = 2V_{pi}/pi$; V_{pi} = 1.5V
E_{MAC} = 5 pJ

Computational efficiency:

Chip	Energy/ MAC	Processor fan-in	Area/MAC (μm^2)	MAC Rate/cm^2
Silicon Photonic (Princeton)	5 pJ	102	20,000	1×10^{14}
Nanophotonics	**7.4 aJ**	**300**	**20**	$\mathbf{1 \times 10^{17}}$
TrueNorth (Electronic)	26 pJ	256	4.9	2×10^{8}

Photodiode Power

Responsivity : Responsivity of a detector is given as the ratio of the generated photocurrent (I) to the amount of optical power (Po) incident on the detector:

$$R = \frac{I}{P_o} = \frac{q\eta}{h\nu} = \frac{q\eta}{hc}\lambda$$

Quantum Efficiency : A detector is not capable of collecting all the photons and convert them to electron-hole pairs. The number of electrons produced per incident photon is defined as the quantum efficiency , which is usually expressed as a percentage

$$\eta = \frac{Rq}{h\nu}$$

$$(1-\eta)P_o = (1 - \frac{Rq}{h\nu})P_o$$

is wavelength dependent, so if we do WDM we may need to model this

$$P_o$$

Consumed Power:

Note: If data is encoded in the power, is simply the value of my tensor

Photodiode Noise

Shot Noise: $I_{sn} = \sqrt{2q(I_P + I_d)\Delta f}$

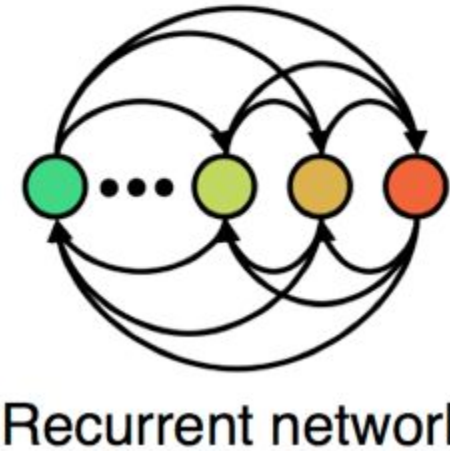
Johnson Noise: $I_{jn} = \sqrt{\frac{4K_B T \Delta f}{R_{sh}}}$

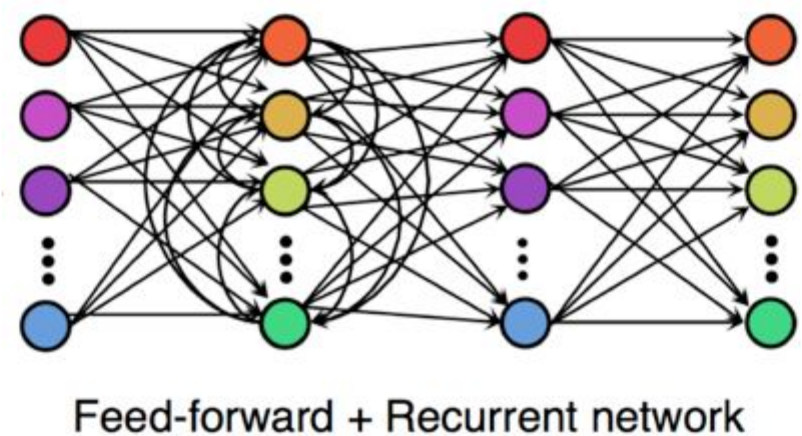
Total Noise Current: $I_{tn} = \sqrt{I_{sn}^2 + I_{jn}^2}$

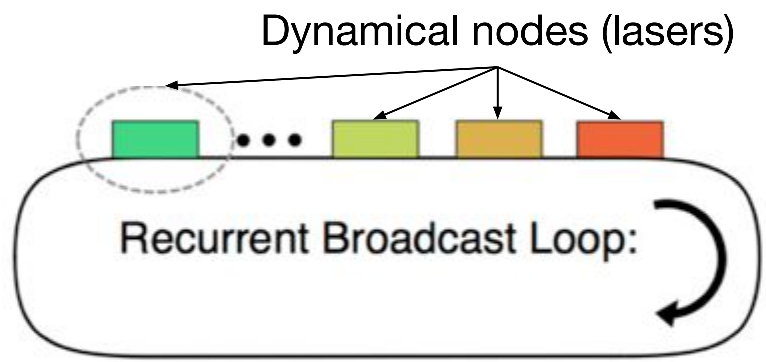
Model with a Gaussian process:

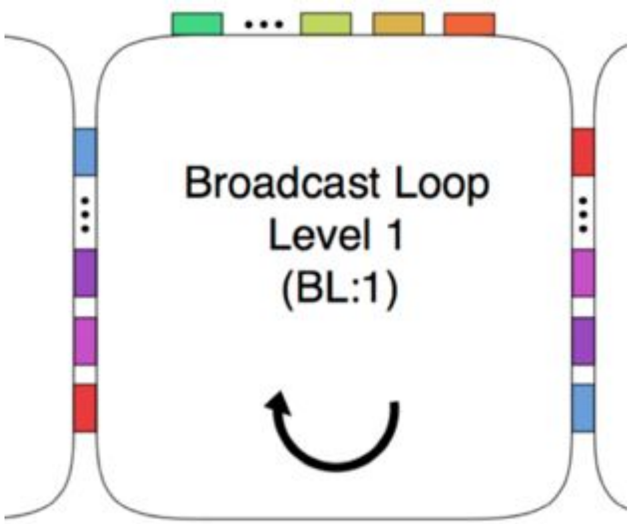
$$\mu = 0$$
$$\sigma = V_{reverse} \times I_{tn}$$

Photonic Implementations



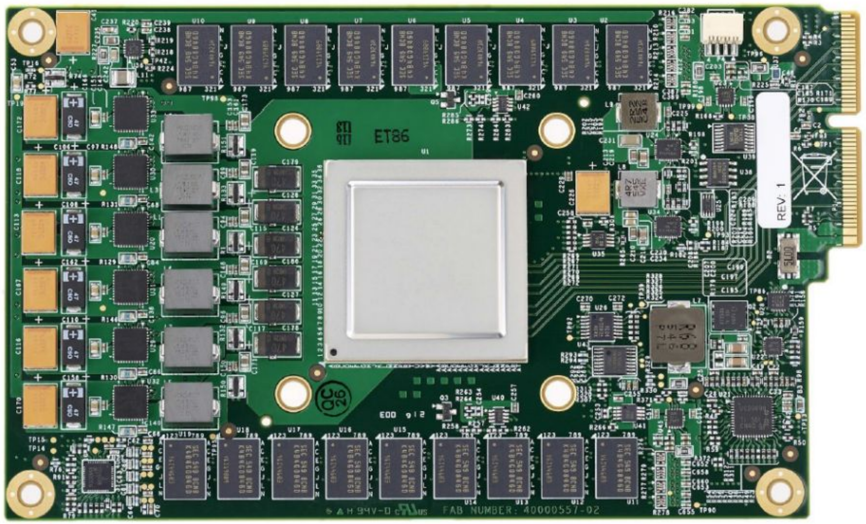


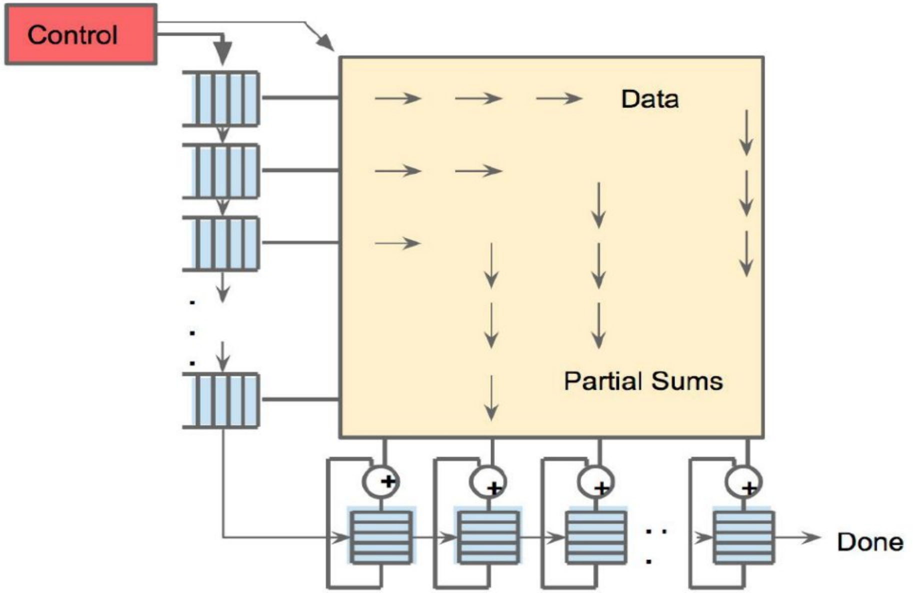




Ferreira de Lima et al. *Nanophotonics* **6** (2017)

Google TPU

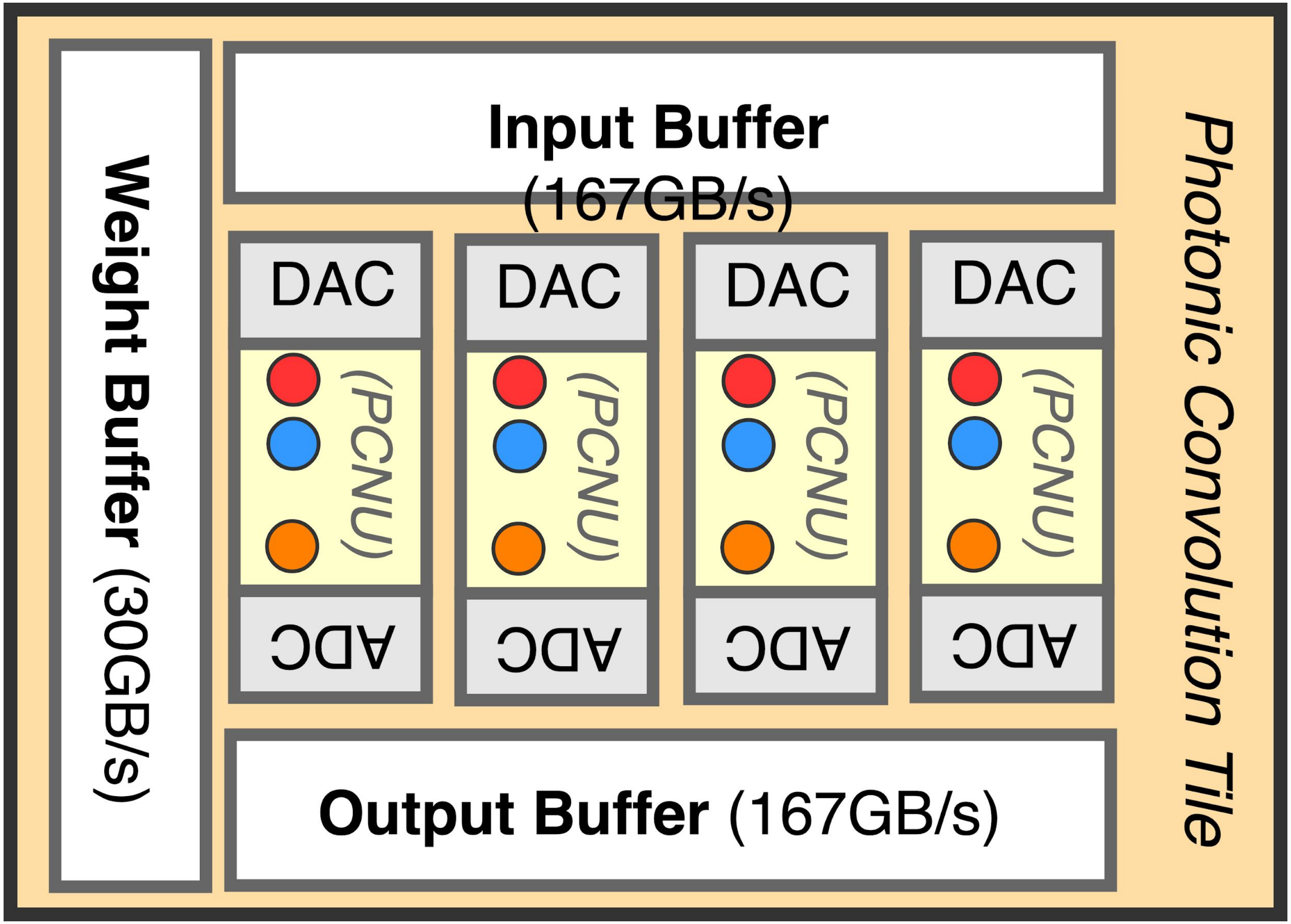




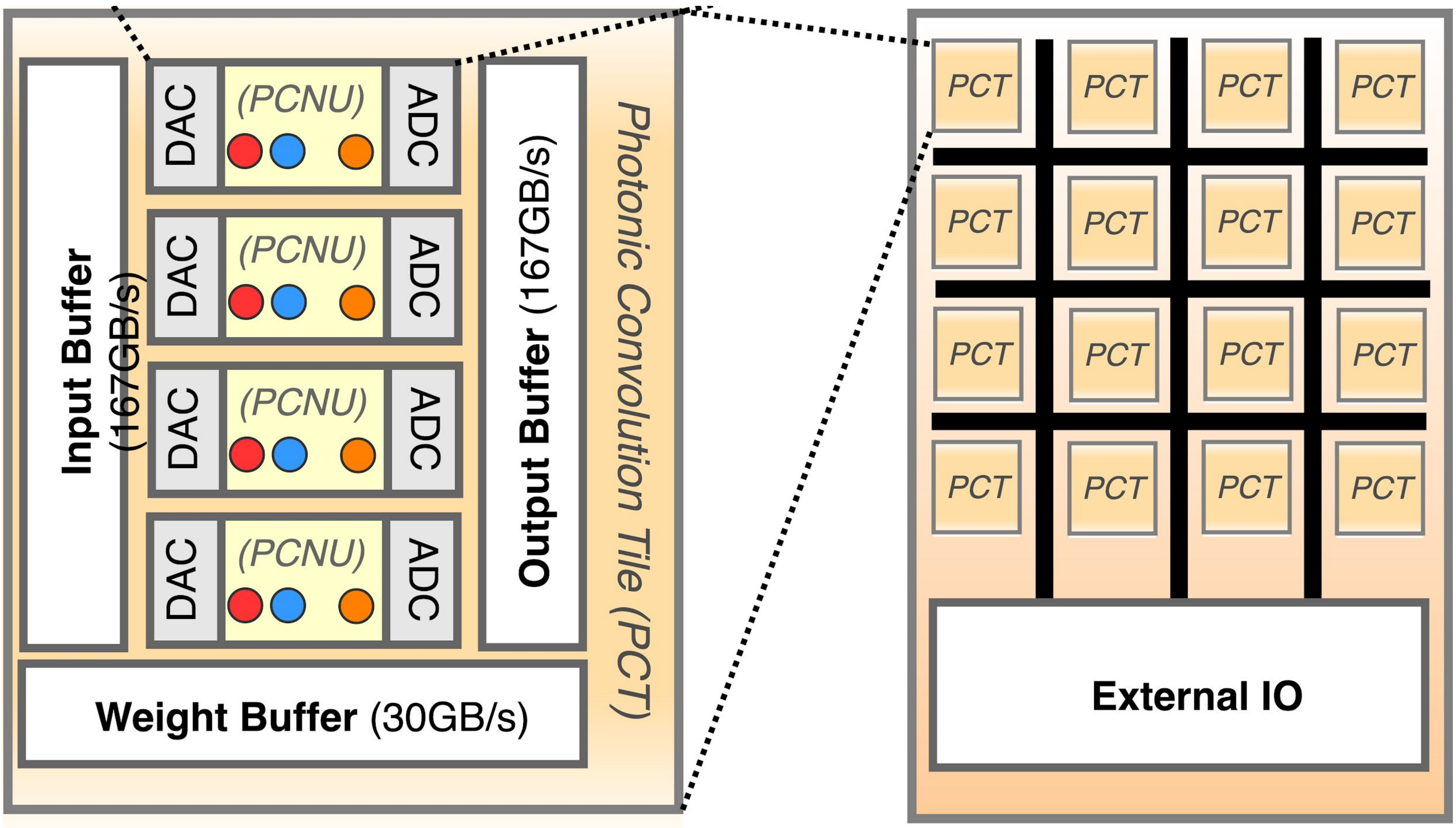
TPU Printed Circuit Board. Systolic data flow of the Matrix Multiply Unit. Software for an SATA disk in a server, but the card uses PCIe Gen3 x16.

Systolic data flow of the Matrix Multiply Unit. Software for an SATA disk in a server, but the card uses PCIe Gen3 x16. has the illusion that each 256B input is read at once, and they instantly update one location of each of 256 accumulator RAMs.

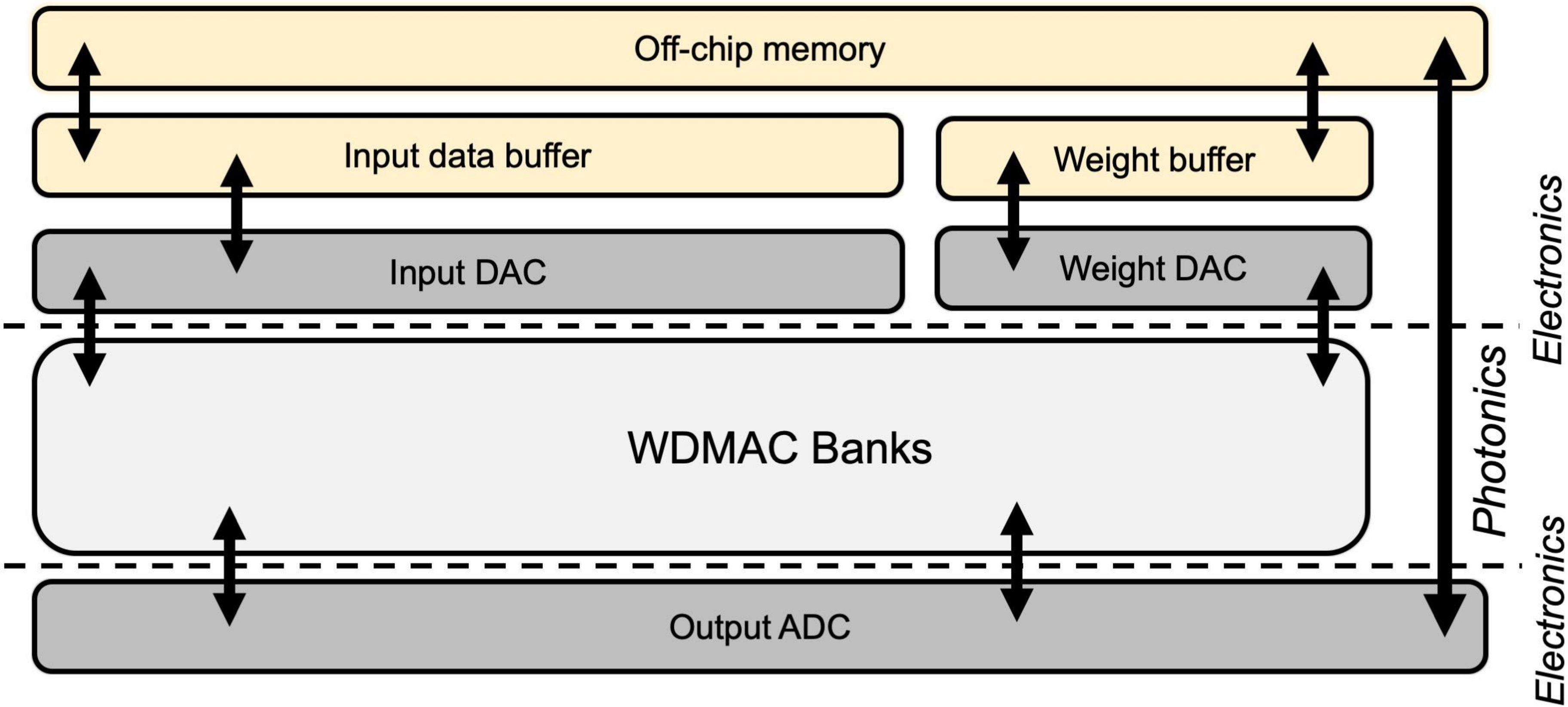
Photonic Convolution Tile



Photonic Convolution Chip



High-Level Architecture



DAC

GW

TABLE I
PERFORMANCE COMPARISON WITH OTHER STATE-OF-THE-ART CMOS DACS

	[1]	[8]	[2]	[9]	**This**
Technology	28 nm	28 nm	65 nm	65 nm	**65 nm**
Supply [V]	1.1, 1.5	2.5, 1.4	1.1, 2.5	1.2	**1.2**
Resolution [Bits]	8	6	6	4	**6**
Architecture	2(U*), 6(B**)	3(U), 3(B)	4(U), 2(B)	4(B)	**6(B) TI*** **
Conversion [GS/s]	18	7	56	10	**20**
Max Signal [GHz]	8.8	3.4	26.9	4.53	**9.2**
Power [mW]	84	145	750	30	**136**
SFDR [dB] (up to Nyquist)	≥38	≥26	≥30	≥23.3	**≥35.1**
SFDR [dB] (Input freq. ≤5GHz)	≥38	≥26	≥42.8	≥23.3	**≥45.5**
Core Area [mm^2]	0.05	0.035	0.24	0.034	**0.072**
FOM [fJ/conv.]	73	1302	537	276	**158**

*U: Unary structure, **B: Binary structure, ***TI: 2x Interleaved DAC

1154

A 65-nm CM[...] DAC With Full-Binary Sub-DACs

Si-Nai Kim, *Member, IEEE*, Woo-Cheol Kim, Min-Jae Seo, and Seung-Tak Ryu, *Senior Member, IEEE*

ADC

GW

IEEE JOURNAL OF SOLID-STATE CIRCUITS, VOL. 52, NO. 4, APRIL 2017 1091

A 23-mW 24-GS/s 6-bit Voltage-Time Hybrid Time-Interleaved ADC in 28-nm CMOS

Benwei Xu, *Student Member, IEEE*, Yuan Zhou, *Student Member, IEEE*, and Yun Chiu, *Senior Member, IEEE*

TABLE I

PERFORMANCE COMPARISON

	This Work	[9]	[7]	[6]	[11]
Architecture	TI-SAR	TI-SAR	TI-BS	TI-flash	TI-SAR
Technology	28nm	32nm SOI	65nm	32nm SOI	28nm SOI
Power Supply (V)	0.85/0.95	0.9/1	1	0.9	1
Sample Rate (GS/s)	24	36	25	20	10
Resolution (bits)	6	6	6	6	6
SNDR/SFDR @ LF (dB)	34.8/53.9	32.6/47	32/42	34.8/NA	34/39
SNDR/SFDR @ HF (dB)	28.9/41	31.6/42.8	29.7/40	30.7/NA	34.3/41.1
Active Area (mm^2)	0.03	0.05	0.24	0.25	0.01
Power (mW)	23	110	88	69.5	32
FoM @ LF/HF (fJ/c-s)	21/42	88/98	108/143	76/124	81/81

TIA

A 40-GHz Mirrored-Cascode Differential Transimpedance Amplifier in 65-nm CMOS

Sang Gyun Kim, Chaerin Hong, *Student Member, IEEE*, Yun Seong Eo, *Member, IEEE*, Jihoon Kim, *Senior Member, IEEE*, and Sung Min Park, *Senior Member, IEEE*

Para[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	**work**
Technology	0.13-µm SiGe	65-nm CMOS	0.13-µm SiGe	45-nm SOI CMOS	65-nm CMOS	0.25-µm SiGe	65-nm CMOS	65-nm CMOS	0.13-µm SiGe	**65-nm CMOS**
Architecture	SF+CB (diff.)	RGC+PA (single)	CB (diff.)	Inverter (single)	DMF (diff.)	RGC (diff.)	RGC (single)	SF (single)	CS+LA (diff.)	**MC (diff.)**
Supply (V)	3.3	3.3/1.0	3.3	1.0	1.2	2.5	1.2	1.6/2.2	3.3	**1.2**
Bandwidth (GHz)	23	21.4	53	30	50	26	21.6	40	29	**40**
TZ gain (dBΩ)	76.5	76.8	80	55	52	53	46.7	55	80	**54**
PD cap. (fF)	65	N/A	N/A	60	50	150	200	40	17	**50**
Noise current spectral density (pA/$\sqrt{\text{Hz}}$)	15.82	17.77	24.86	20.47	22.42	21.3	30	12.5	50*	**19.8**
GD variation (ps)	N/A	N/A	N/A	±3.9	N/A	±9.5	±8 (sim.)	±32	N/A	**±10**
Power diss. (mW)	67.5	137.5	277	9	49.2	36.4	39.9	122	77	**55.2**
Chip size (mm^2)	1.24	0.32	1.95(dual)	0.29	0.96	0.75	0.56	0.54	0.52	**0.6**

SF = source follower, CB = common-base, RGC = regulated cascode, PA = post-amplifier, DMF = dual-mode feedforward, CS = common-source
* calculated from the measured sensitivity

AIM Photonics Passive Components

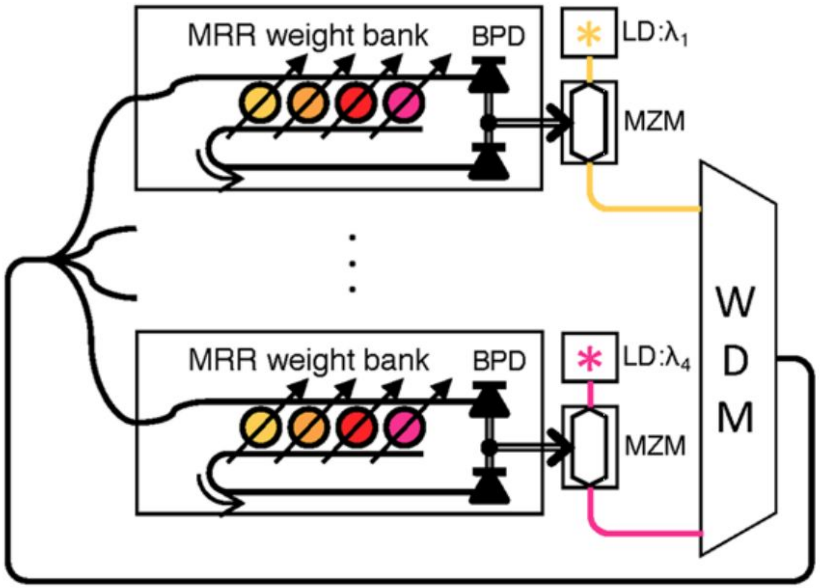
Passive Components	Qty	Selected Performance
O+C+L Band Si & SiN waveguides	6	Low Loss Si and SiN <1dB/cm
C+L Band Edge Couplers	2	<2.8dB/facet (SMF28 -TE only), <1.8dB/facet (UHNA7 fiber – TE/TM)
O Band Edge Couplers	2	<1.8dB/facet (UHNA1 fiber – TE/TM)
Vertical Couplers (Si & SiN)	2	<2.8dB loss
3dB 4-Port Couplers (Si & SiN)	2	loss ~0.5dB, split ratio deviation <0.8% std
O+C+L Band Y- Junctions (Si & SiN)	3	loss ~0.35dB, split ratio deviation <0.6% std
Power Taps (1% & 10%)	3	loss <0.1dB, deviation <5% of target tap ratio
Layer Transitions, Escalators	3	loss <0.1dB
Crossing (Si)	1	loss <0.2dB, crosstalk <-45dB
Polarization Rotator	1	loss <0.8dB
Polarization Splitter and Rotator	1	loss <0.8dB, PDL<0.3dB

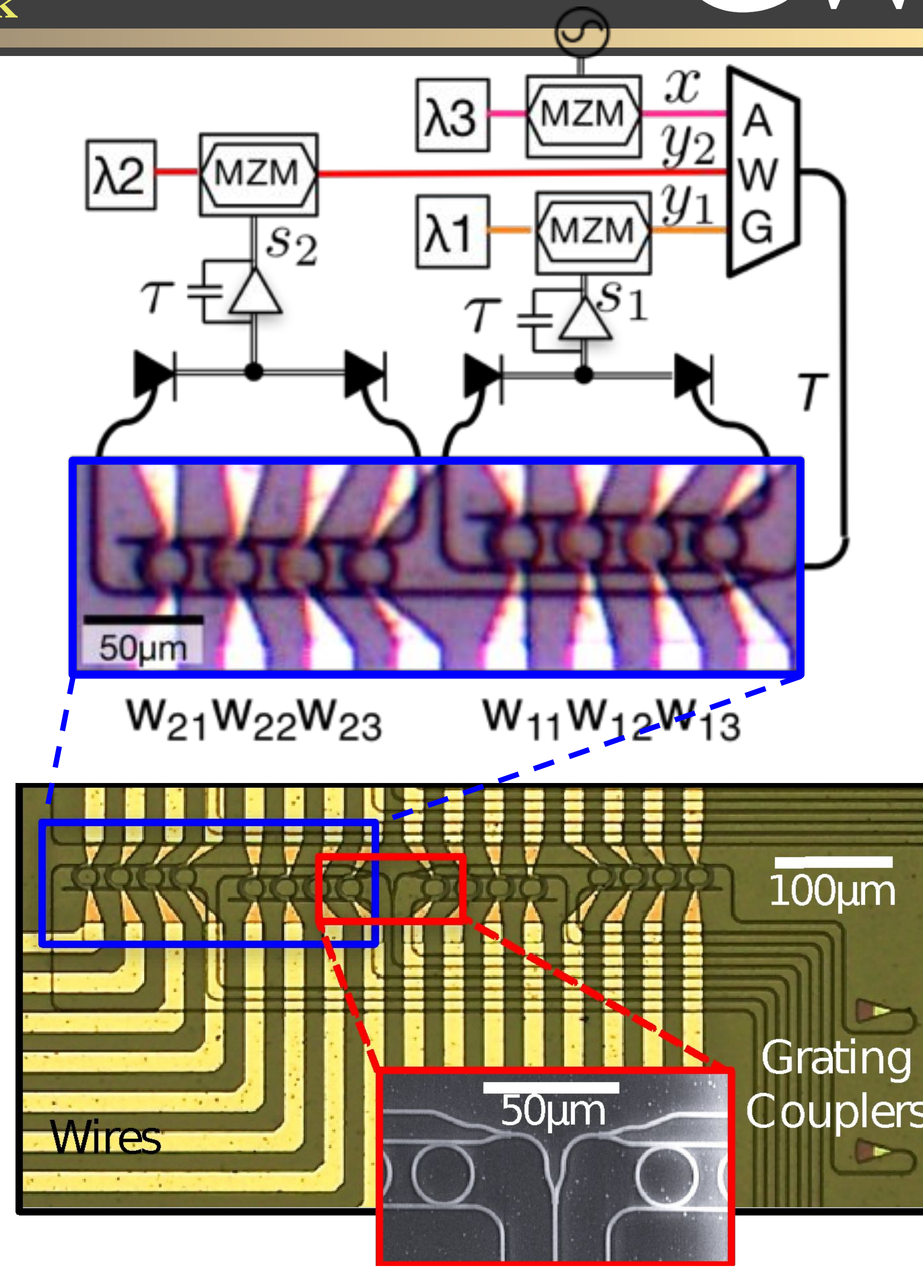
AIM Photonics Active Components

Active Devices	Qty	Selected Performance
Digital C Band Photodetector	1	BW>45GHz, ~15nA Dark Current, R~1A/W
Digital C+L Band Photodetector	1	BW>35GHz, R~1.1A/W
Digital O Band Photodetector	1	BW>40GHz, ~11nA Dark Current, R~0.85A/W
Digital C+L Band MZM	1	50Gbaud (100Gbps) PAM4 2Vpp differential
Digital O Band MZM	1	25Gbaud (50Gbps) PAM4 2Vpp differential
Microring Filters (Si, Tunable)	4	<0.15dB loss, ~26nm FSR, >1nm/mW tuning
Microdisk Switches (Si, Tunable)	4	<3ns (10%-to-90%) switch time, <2 dB loss, ~1V
O+C+L Band Microdisk Modulators (Si, Tunable)	5	25Gbaud (50Gbps) PAM4, ~1Vpp, >4dB ER
Analog C Band Photodetector	1	BW~35GHz, ~60nA Dark Current, R~0.9A/W SFDR>113dB/Hz$^{2/3}$
Analog C+L Band MZM	1	BW>15GHz, SFDR>100dB/Hz$^{2/3}$
Thermo-Optic Phase Shifter (Si)	2	<0.15dB loss, <35mW/π
Thermo-Optic Switch (Si)	2	<0.15dB loss, <50mW/π
Variable Optical Attenuator	1	up to 10dB controlled attenuation

Related Work

Small (4-node) Recurrent Neural Network





Continuous-Time Recurrent Network Model

Weight matrix **State vector** **Weighted input**

$$\frac{d\vec{s}(t)}{dt} = \mathbf{W}\vec{y}(t) - \frac{\vec{s}(t)}{\tau} + \vec{w}_{in}x(t)$$

$$\vec{y}(t) = \sigma\left[\vec{s}(t)\right]$$

Time-constant

output **Nonlinear function**

Tait, Alexander N., et al. "Neuromorphic photonic networks using silicon photonic weight banks." Scientific reports 7.1 (2017): 1-10.

The case for optical brain simulator systems

- Full brain simulation is a scientific dream yet to be realized
- Some project such as Spinnaker-2 attempt to tackle this challenge
- However, even those efforts target full brain simulation with **simpler neuron** and **synaptic models**
- Brain simulation problem is communication bound [Moore et al, 2012]
- Nerves system has short range and long-range synaptic connections Optics offer energy efficient communications
- Neurons in the brain have large fan-in, fan-out WDM
- Even processing can be done using photonics (Princeton spiking neurons) at a much lower energy budget